Calibrating the EQ-SANS chopper phase offsets from time-of-flight spectra

A two-flight-path slit method for the six-disc system

Changwoo Do
Neutron Scattering Division, Oak Ridge National Laboratory

Draft, November 12, 2026

Abstract

Since the 2026B cycle EQ-SANS has six chopper discs in three pairs. To set a wavelength band, the control system converts the requested wavelengths into a phase for every disc. That conversion needs one calibration constant per disc, the *phase offset*. We know where the discs are and how wide their openings are, but not their offsets. The offsets have to come from measured time-of-flight (TOF) spectra.

This report explains, step by step, why the present way of extracting the offsets is biased. The present method reads one chopper edge in the beam-monitor spectrum and assumes that neutrons leave the moderator at the instant of the proton pulse. The bias is $\bar{t}_0\,(1 - L_i/L_{\mathrm{mon}})$, where $\bar{t}_0 \approx 120\,\mu$s is the mean emission delay of the moderator. In a simulated instrument this method returns offsets that are 12–94 μs too large. When those offsets are used, every disc edge lands 0.03–0.05 Å too short.

We then propose a method that needs no reference sample and no knowledge of the emission delay or of the DAS time zero. The two discs of a pair are used together as a narrow slit, and the slit is recorded at two flight paths at once (beam monitor and detector). A slit-width scan then separates the two discs of each pair. On the same simulated instrument the offsets come out within 3.5 μs and the edges within 0.005 Å. When the instrument model is deliberately made wrong (a different emission-time shape, a different beam sweep), they stay within 8 μs and 0.01 Å. Every number in this report is produced by the scripts listed in Appendix C.

Contents

1 Why the phase offsets matter **3**

2 Reading a time–distance diagram **3**

3 A disc is a gate with sloped sides **4**

4 What we do not know **5**

4.1 The six phase offsets δ_i . . . 5
4.2 When neutrons leave the moderator . . . 5
4.3 The DAS time zero . . . 6
4.4 The equations . . . 6
4.5 What can and cannot be determined . . . 6

5 The present method and why it is biased **7**
5.1 What it does . . . 7
5.2 Where the bias comes from . . . 7
5.3 What it does to the band . . . 8
5.4 A sign in the existing records . . . 8

6 The proposed method **9**
6.1 A pair of discs makes a slit . . . 9
6.2 Two flight paths pin down the neutron path . . . 9
6.3 Scanning the slit across wavelengths . . . 10
6.4 What can go wrong with the slit scan alone: the beam sweep . . . 11
6.5 Separating disc a from disc b: the width scan . . . 12
6.6 The instrument model and the fit . . . 13

7 One measurement, step by step **13**

8 Testing on a simulated instrument **16**
8.1 Set-up . . . 16
8.2 Results . . . 17
8.3 What if the model is wrong? . . . 18
8.4 Reading the results . . . 19

9 How to do it on the instrument **20**
9.1 Do we need to know the current offsets? . . . 20
9.2 Set-up . . . 21
9.3 What to type: the worksheet . . . 21
9.4 The scan script . . . 21
9.5 Option: spin only the pair under test . . . 23
9.6 Run list . . . 23
9.7 Analysis . . . 23
9.8 Keeping it calibrated . . . 24

10 The analysis tool: `slitcal` **24**

11 Limitations and open points **28**

A Worksheet for the instrument as of 12 November 2026 **29**
A.1 Default: all six discs spinning . . . 29
A.2 Option: only the pair under test spins . . . 30

B Derivations **31**
B.1 Bias of the single-edge method and its effect on the band . . . 31
B.2 The area theorem . . . 31

C Software and reproduction **32**

D Notation **32**

1 Why the phase offsets matter

EQ-SANS is a time-of-flight instrument. Every proton pulse, 60 times a second, produces a burst of neutrons with all wavelengths. Bandwidth choppers, spinning discs with a cut-out, decide which wavelengths go on to the sample. A neutron passes a disc only if it arrives while the cut-out is in the beam. Slow (long-wavelength) and fast (short-wavelength) neutrons arrive at different times, so the timing of each disc selects wavelengths.

The control system (the EPICS IOC `bl6-SkfChopper`, routine `calc6Phases` in `hexaSub.c`) turns a requested band into one number per disc, the `TotalDelay`. This is the time after the proton pulse at which the centre of the disc's cut-out should cross the beam. The hardware cannot produce that time exactly as written. The angle of the encoder mark on the disc, the cable delays and the controller's own latency all add an unknown but constant time. That constant is the **phase offset** δ_i of disc i:

$$t_i^{\text{centre}} \;=\; P_i - \delta_i \pmod{16\,667\ \mu\text{s}}, \tag{1}$$

where P_i is the commanded `TotalDelay` and t_i^{centre} is when the centre of the cut-out actually crosses the beam.

How much does an offset error matter? A neutron of wavelength λ needs $L\lambda/(h/m_n)$ to fly a distance L, where $h/m_n = 3956\,\text{m}\,\text{Å/s}$. So an error Δt in the timing of a disc at distance L moves the wavelength at which that disc opens or closes by

$$\Delta\lambda = \frac{h}{m_n}\,\frac{\Delta t}{L}. \tag{2}$$

At station 1 ($L \approx 5.67\,\text{m}$) an error of $10\,\mu\text{s}$ shifts an edge by $0.007\,\text{Å}$; an error of $100\,\mu\text{s}$ shifts it by $0.07\,\text{Å}$. For the ordinary white-beam band of about $4\,\text{Å}$ this hardly matters. For the new monochromatic mode it matters a great deal. A 5 % band at $3\,\text{Å}$ is only $0.15\,\text{Å}$ wide.

Table 1: The six discs (finalized values, drtsans `EQSANS_chopper_configurations.json`, entry 2026-03-04). The "a" disc of a pair is the upstream one. "EPICS" is the 1-based chopper index used by the IOC.

Disc	EPICS index	Distance from moderator (m)	Cut-out (deg)	Open time at 60 Hz (μs)
1a	5	5.6601247	129.600	6000
1b	1	5.6757668	129.600	6000
2a	6	7.7601247	180.000	8333
2b	2	7.7757668	180.000	8333
3a	3	9.4978	230.010	10648
3b	4	9.5078	230.007	10648

The question this report answers is the following. *Assume we know only the distances and opening angles in Table 1. How do we determine the six offsets δ_i accurately from TOF spectra recorded on the instrument?*

2 Reading a time–distance diagram

All the arguments below use one kind of picture, so we start there (Fig. 1). The horizontal axis is time after the proton pulse, called T0. The vertical axis is distance from the moderator. A neutron moving at constant speed v traces a straight line; its slope, distance over time, is the speed:

$$L = v\,(t - t_0), \qquad v = \frac{h}{m_n\lambda}, \tag{3}$$

where t_0 is the moment it leaves the moderator. Fast, short-wavelength neutrons have steep lines. Slow, long-wavelength neutrons have shallow lines. A vertical line would mean infinite speed, so it never occurs. The horizontal grey lines mark where things are: the three chopper stations, the beam monitor at 10.006 m, the sample at 14.12 m and the detector, 18.12 m from the moderator for a 4 m sample–detector distance.

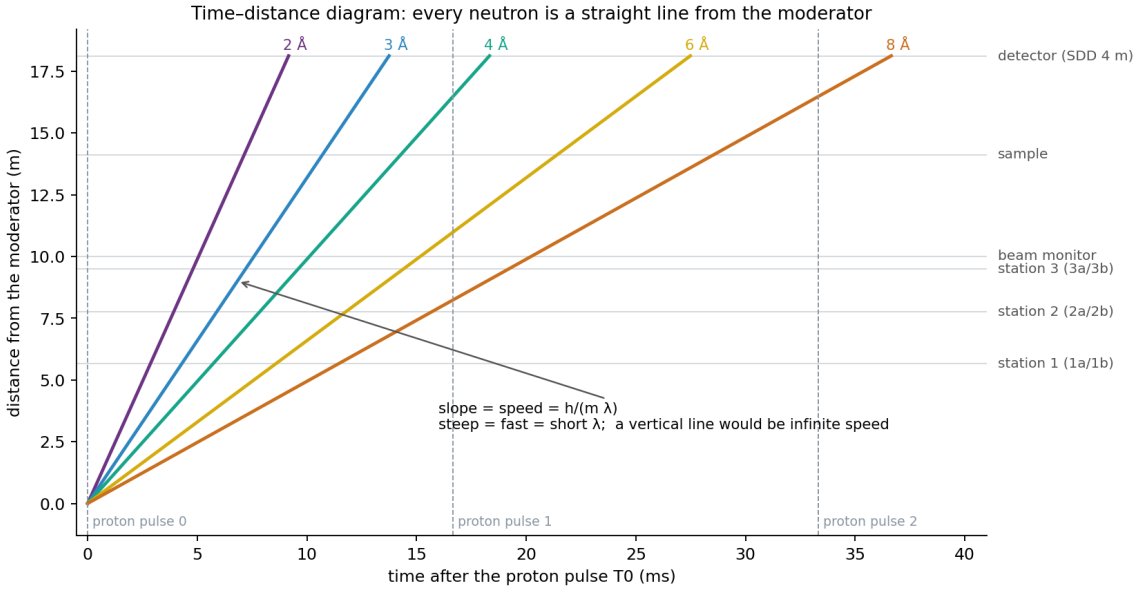


Figure 1: Time–distance diagram. Each coloured line is one wavelength leaving the moderator at T0. A chopper is a horizontal line that is "open" only during certain time intervals. A neutron reaches the detector only if its line crosses every station while that station is open.

Two facts from this picture carry the whole report.

- **Fact 1.** For a fixed wavelength, the time at which a neutron reaches any distance is fixed once its departure time t_0 is fixed. If we know where a line is at two distances, we know the whole line: its wavelength (slope), when it passed every disc, and when it left the moderator.
- **Fact 2.** Neutrons do not all leave at T0. They leave over a few hundred microseconds (Section 4.2). On the scale of Fig. 1 (milliseconds) that is invisible. On the scale of a phase offset (microseconds) it is everything.

3 A disc is a gate with sloped sides

Figure 2 shows one disc and its transmission as a function of the time a neutron arrives. The cut-out of a station-1 disc spans 129.6°; at 60 Hz it is in the beam for $129.6/360 \times 16\,667 = 6000\,\mu$s.

The beam is not a line. It is about 40 mm wide (the value written in the comments of `hexaSub.c`) and sits $r \approx 289$ mm from the disc axis. The edge of the cut-out moves at $2\pi r f = 109$ m/s. It therefore needs

$$\tau_b = \frac{w_{\text{beam}}}{2\pi r f} = \frac{0.040}{2\pi \times 0.289 \times 60}\ \text{s} \approx 367\ \mu\text{s} \tag{4}$$

to sweep across the beam. During that time the transmission rises, or falls, roughly linearly. We define the *edge time* as the 50 % point, the moment the edge crosses the centre of the beam. The

gap centre (OGC) is the point midway between the two edges. It is the quantity that `TotalDelay` sets.

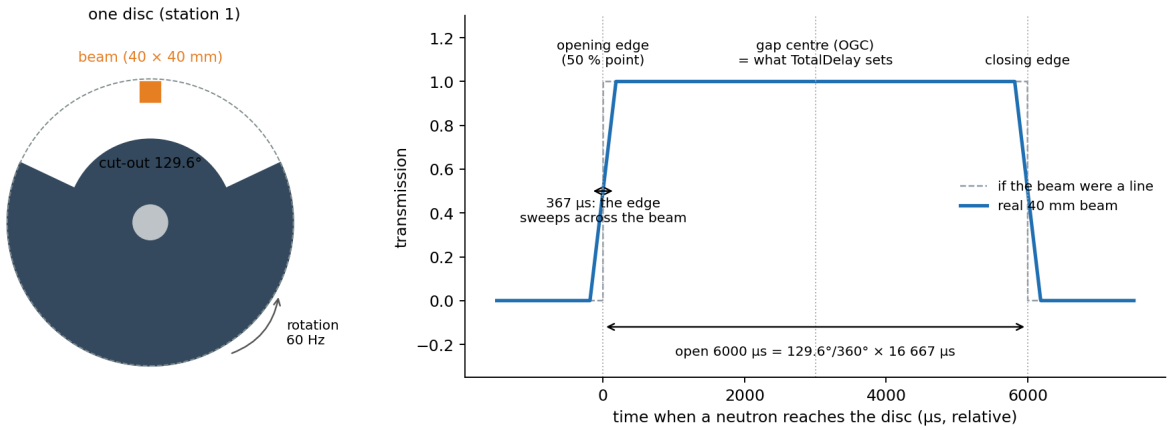


Figure 2: Left: a station-1 disc with its 129.6° cut-out and the beam. Right: transmission versus arrival time. With a finite beam each edge is a 367 μs ramp (blue) rather than a step (grey). The edge time is the 50 % point; the gap centre lies midway between the two edges.

4 What we do not know

Three things enter the TOF spectra that we do not know in advance.

4.1 The six phase offsets δ_i

These are what we want (Eq. 1). They are about 15 000 μs at 60 Hz, and the values currently in use are listed in Table 2 for reference.

Table 2: Phase offsets currently in drtsans (60 Hz; the 30 Hz values are twice these). The IOC has the same values except 14565.6 for 3b.

Disc	1a	1b	2a	2b	3a	3b
δ_i (μs)	15072.89	14954.46	14834.04	14805.40	14726.06	14565.69

4.2 When neutrons leave the moderator

A neutron does not leave the moderator at the instant the protons hit the target. It has to slow down in the hydrogen first, and that takes time. The departure time t_0 is spread out. Its distribution depends on wavelength and on the moderator, and it is not something we can read off a datasheet.

What we do know comes from drtsans (Fig. 3, left).

- `emission_delay`(λ) gives the *mean* delay $\bar{t}_0$. It is about 120 μs at 2 Å, rising slowly to 130 μs at 16 Å. drtsans subtracts it from every measured TOF before converting to wavelength.
- `moderator_time_uncertainty`(λ) is exactly twice the mean, about 250 μs. drtsans uses it as a width in the resolution calculation.

drtsans does *not* give the shape of the distribution. In this report we assume a Gamma distribution with shape $k = 2$ and the drtsans mean (Fig. 3, right). It starts at zero, peaks at half the mean and has a tail. With that assumption 99 % of the neutrons have left by about $3.3\,\bar{t}_0 \approx 400\,\mu$s. That is where the "0–400 μs" in our figures comes from. **It is a consequence of the assumed shape, not a measurement.** The shape is the least certain ingredient of the model. Section 8.3 therefore repeats the whole test with a very different shape (an exponential, $k = 1$) and shows that the proposed method does not depend on it.

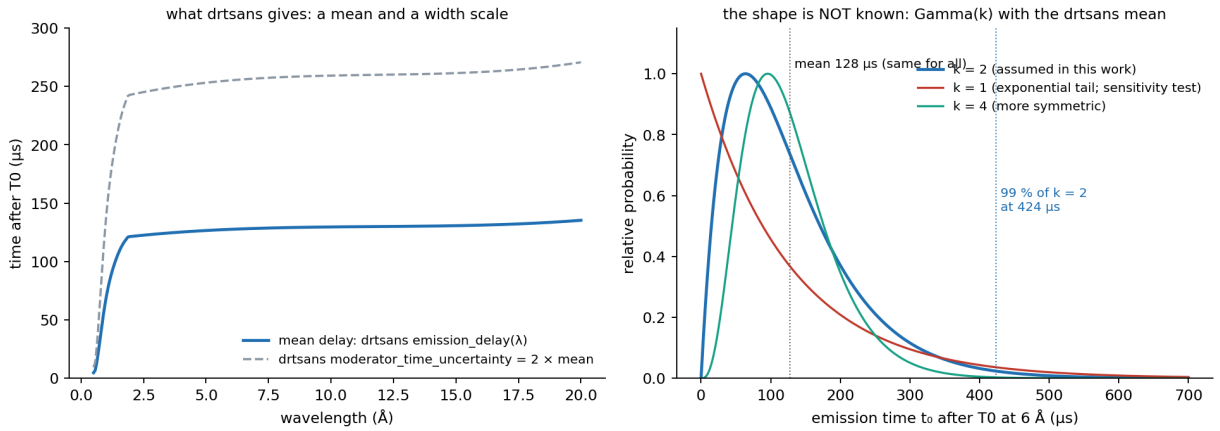


Figure 3: Left: what drtsans provides, a mean emission delay and a width scale twice as large. Right: the distribution of departure times at 6 Å for three Gamma shapes that share the drtsans mean. $k = 2$ is used throughout this report; $k = 1$ is used to test robustness.

4.3 The DAS time zero

The data-acquisition system stamps every detected neutron with its TOF, the time since a T0 signal. That signal may be late or early by some d_0 relative to the real proton pulse. Nobody has measured d_0 for EQ-SANS, and it is presumably a few microseconds.

4.4 The equations

With these three unknowns, a neutron of wavelength λ that leaves at t_0 crosses disc i (distance L_i) at

$$t_i = t_0 + \frac{L_i \lambda}{h/m_n}, \tag{5}$$

and is recorded at a counter at distance L_x with the TOF

$$T_x = t_0 + \frac{L_x \lambda}{h/m_n} + d_0. \tag{6}$$

It passes disc i only if t_i falls inside the open window of that disc, whose centre is $P_i - \delta_i$.

4.5 What can and cannot be determined

There is one combination that no TOF measurement can ever reveal. Suppose we add the same amount ε to every offset δ_i (every disc really opens ε earlier), subtract ε from every departure time, and add ε to d_0. Then:

- every neutron leaves ε earlier and still reaches every disc while it is open, because the discs also moved ε earlier;
- every recorded TOF is unchanged, because the earlier departure ($-\varepsilon$) cancels the later time stamp ($+\varepsilon$).

So $(\delta_i + \varepsilon,\ \bar{t}_0 - \varepsilon,\ d_0 + \varepsilon)$ looks exactly like $(\delta_i, \bar{t}_0, d_0)$.

Fortunately, nothing we need depends on ε. Operating the instrument needs only two combinations, and both are unchanged by the shift.

1. $\delta_i + \bar{t}_0$: *which wavelength a disc passes for a given command.* The disc edge meets neutrons that left at $\bar{t}_0$, so the edge wavelength depends on (disc time) − (departure time).
2. $\bar{t}_0 + d_0$: *how to turn a measured TOF into a wavelength.* From Eq. 6, $\lambda = (h/m_n)(T_x - \bar{t}_0 - d_0)/L_x$.

We may therefore fix $d_0 = 0$ in any analysis, as a convention, without losing anything. The fitted offsets and the fitted emission delay then each absorb part of the true d_0, but the two combinations come out right.

Why one flight path is not enough. With only the monitor, each disc's edge appears at one TOF, Eq. 6. To turn that TOF into a disc time we must know when the neutrons left ($\bar{t}_0 + d_0$). With one counter there is no way to measure that: each disc's own offset can absorb any error in it. The present method resolves this by assuming $\bar{t}_0 = d_0 = 0$. That assumption is the source of its bias (Section 5). With two counters at different distances the same neutrons give two TOFs. By Fact 1 that fixes their line completely, including when they left.

5 The present method and why it is biased

5.1 What it does

The tools in `/SNS/EQSANS/shared/script/choppers/` (`find_edges_ui.py`) work as follows.

1. Spin one disc at a known command P_i; leave the others parked open.
2. Record the beam-monitor spectrum. The disc's cut-out shows up as a plateau with a rising and a falling edge.
3. Fit each edge with an error function between two flat levels. This gives the edge TOF T at the monitor.
4. Convert T to the time the edge was at the disc, $x = L_i\,T/L_{\mathrm{mon}}$. Then OGC = $x\pm$ half the open time, and $\delta_i = P_i - \mathrm{OGC}$.

5.2 Where the bias comes from

Step 4 scales the TOF by L_i/L_{mon}. That is the same as drawing a straight line from the origin of the time–distance diagram (departure at exactly T0) to the measured point at the monitor, and reading where it crosses the disc (red dashed line in Fig. 4). The neutrons that make the edge actually left at about $\bar{t}_0$ (blue line). The two lines meet at the monitor and separate towards the moderator. At the disc they differ by

$$x_{\mathrm{true}} - x_{\mathrm{assumed}} = \bar{t}_0 \left(1 - \frac{L_i}{L_{\mathrm{mon}}}\right) \tag{7}$$

(derivation in Appendix B.1). With $\bar{t}_0 = 120\,\mu$s this is 52 μs for station 1, 27 μs for station 2 and 6 μs for station 3. The disc time is read too early, so the offset comes out too large by the same amount.

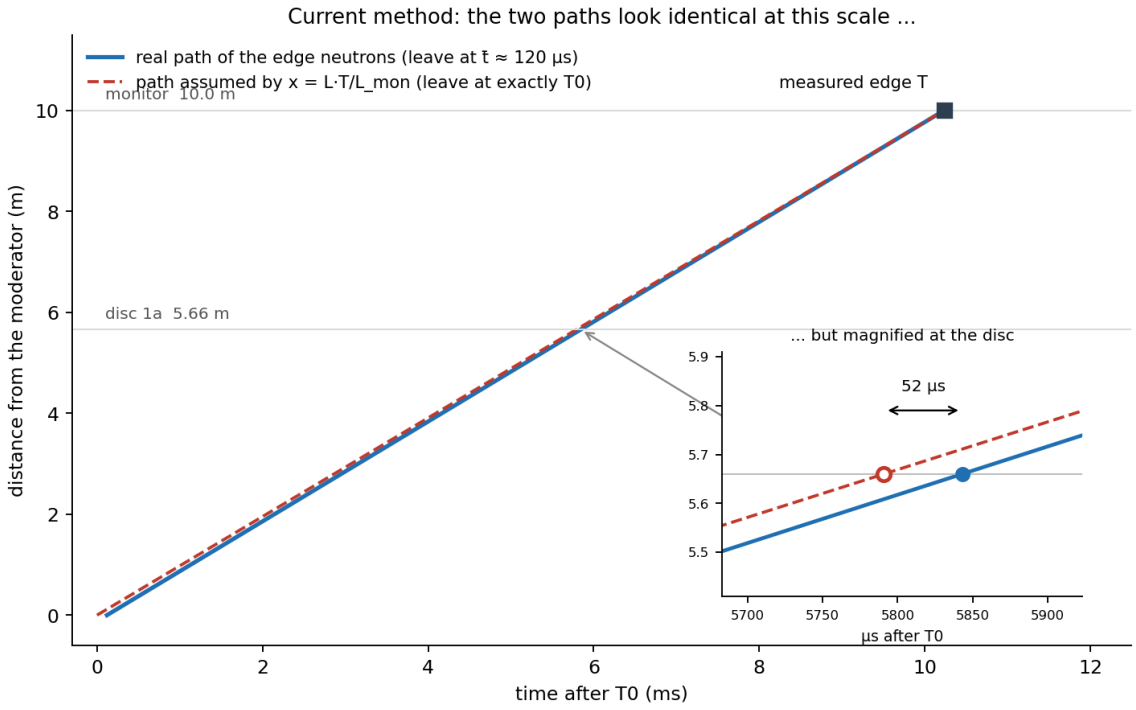


Figure 4: The present method in a real time–distance diagram. On the millisecond scale the assumed path (red, leaving at T0) and the real path (blue, leaving at $\bar{t}_0$) are indistinguishable. Magnified at the disc (inset), they cross the disc 52 μs apart. Both axes of the inset are real time and distance.

There is a second, smaller effect. The error-function model assumes the spectrum is flat on both sides of the edge, but the spectrum is sloped. The edge is also asymmetric, because the emission tail and the beam sweep blur it unevenly. Both shift the fitted midpoint by some tens of microseconds, by an amount that depends on the wavelength and on which edge is fitted.

5.3 What it does to the band

Here is a slightly surprising consequence. The offsets from this method are biased by different amounts for different discs. Yet when they are used in `calc6Phases`, which also assumes departure at T0, every disc edge ends up displaced by *the same* wavelength:

$$\Delta\lambda_{\mathrm{edge}} \approx -\frac{h}{m_n}\,\frac{\bar{t}_0}{L_{\mathrm{mon}}} \approx -0.05\ \text{Å} \tag{8}$$

(Appendix B.1). The whole band is shifted to shorter wavelength by about 0.05 Å. This agrees in sign and size with the observation, recorded in August 2026, that drtsans reports a slightly smaller centre wavelength than the requested one in monochromatic mode.

5.4 A sign in the existing records

The notes in `fit_summary.md` (March 2026) list two measurements for several discs, at two phase delays. They differ by 20–300 μs for the same disc (for example 14660 and 14357 μs for 3b). They

also differ by 90–420 μs from the finalized offsets. Those numbers were read by eye ("fall bottom", "rise top"), so they are not a clean test. Still, they show how sensitive a single-edge, single-distance procedure is to choices that the method itself does not constrain.

6 The proposed method

The idea has three parts.

1. Turn each pair of discs into a short slit in time.
2. Record what passes the slit at two distances at once.
3. Separate the two discs of the pair with a width scan.

Finally, fit everything with one model of the instrument.

6.1 A pair of discs makes a slit

The 2026 upgrade put two discs, 15.6 mm apart, at stations 1 and 2; station 3 already had two. If disc "a" of a pair opens at some time and disc "b" closes w microseconds later, neutrons can pass only during those w microseconds. Each disc on its own is open for thousands of microseconds; together they make a slit of any width we like (Fig. 5). The front edge of the slit belongs to disc a and the back edge to disc b. This was not possible at stations 1 and 2 before 2026.

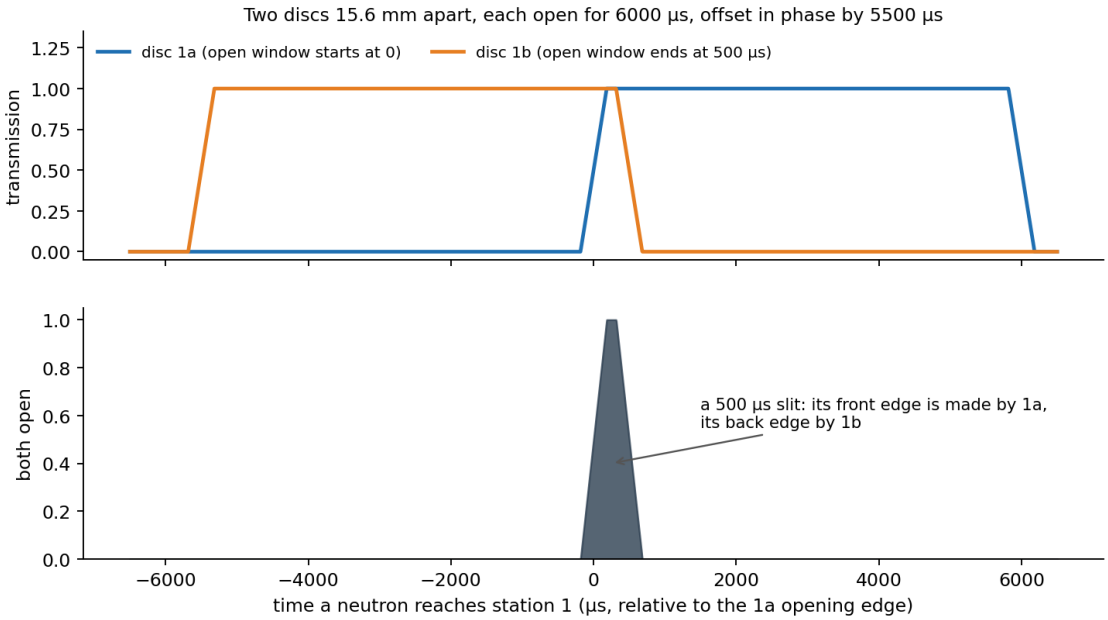


Figure 5: Top: the open windows of discs 1a and 1b, phased so that 1b closes 500 μs after 1a opens. Bottom: the product, a 500 μs slit. Its sloped sides are the 367 μs beam sweeps of the two discs.

6.2 Two flight paths pin down the neutron path

The neutrons that pass a short slit form a small bundle. In a time–distance diagram they fan out from the slit (Fig. 6). The bundle produces a TOF peak at the beam monitor (10.0 m) and another

at the detector (18.1 m), recorded in the same run. The detector sees the direct beam through an attenuator. Take the average neutron, the centroid of each peak. By Fact 1 the two peak times T_m and T_d fix its line, and everything follows from Eqs. 5–6:

$$\lambda = \frac{h}{m_n}\,\frac{T_d - T_m}{L_{\mathrm{det}} - L_{\mathrm{mon}}} \qquad \text{(slope; no } t_0\text{, no } d_0 \text{ needed)} \tag{9}$$

$$t_i + d_0 = T_m - \frac{(L_{\mathrm{mon}} - L_i)\lambda}{h/m_n} \qquad \text{(when the slit was open)} \tag{10}$$

$$\bar{t}_0 + d_0 = T_m - \frac{L_{\mathrm{mon}}\lambda}{h/m_n} \qquad \text{(when the neutrons left)} \tag{11}$$

Subtracting the last two gives $t_i - \bar{t}_0$, the time between departure and the slit. That is exactly the combination needed to set the disc (Section 4.5). The unknown d_0 cancels, and so does any assumption about when neutrons leave. The offset follows from Eq. 1, with t_i the slit centre.

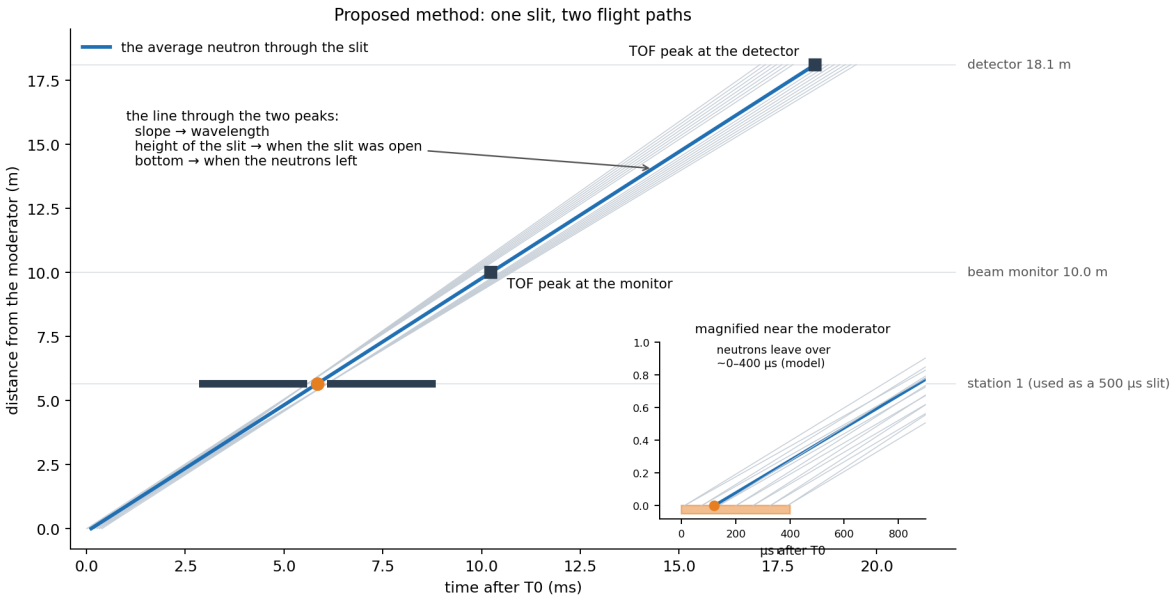


Figure 6: The proposed geometry in a real time–distance diagram. Grey lines: neutrons leaving at different times (0–400 μs) and passing the front or back of a 500 μs slit at station 1. Blue: the average path, fixed by the monitor and detector peaks (squares). Inset: magnified near the moderator, where the departure times differ.

6.3 Scanning the slit across wavelengths

One slit setting calibrates one pair at one wavelength. Moving the slit to different times selects different wavelengths. We use six settings between 3 and 13.5 Å for each of the three pairs. This does three things.

- It averages down the statistical error.
- It measures how the mean emission delay changes with wavelength. drtsans says by about 10 μs over this range; the fit includes a scale factor on the drtsans curve.
- It provides a check on the geometry (Fig. 7). When the slit moves by Δt, the peak at a counter at L_x moves by $(L_x/L)\,\Delta t$, where L is the distance of the slit. For the peak centroid L is the

mean distance of the pair. In the forward model the slope is 1.7650 at the monitor, against $L_{\mathrm{mon}}/\bar{L} = 1.765$. The measured ratio therefore checks the distances. Following the *front* edge of the peak instead of its centroid gives the distance of the opening disc alone. That tells whether it is 1a or 1b: their distances differ by 0.28 %, which over a 5 ms scan is about 45 μs at the detector, easily seen with edges located to a few microseconds. This settles the a/b naming question directly from data.

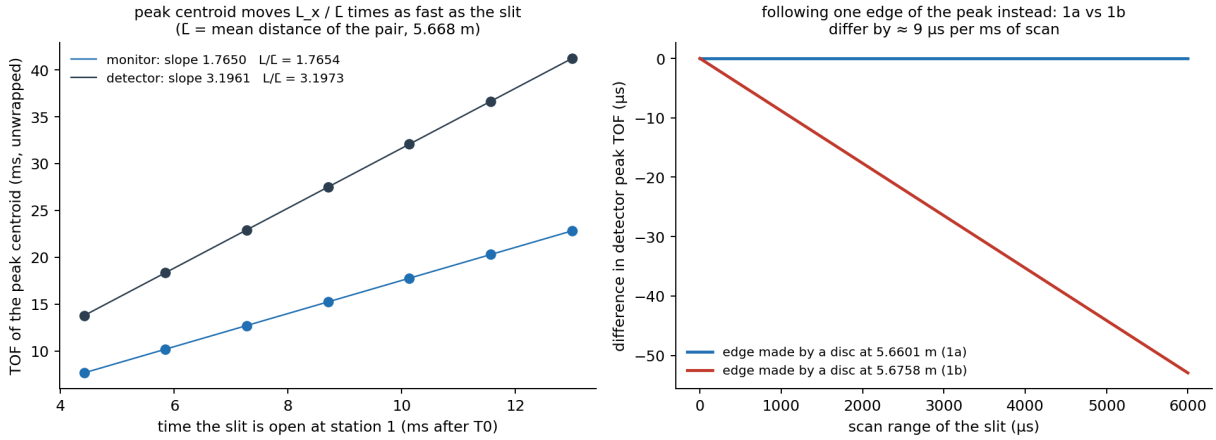


Figure 7: Left: the TOF of the peak centroid at the monitor and detector versus the time the slit is open (forward model, station 1). The slopes equal L_x divided by the mean distance of the pair. Right: the difference in detector peak position between assuming disc 1a or 1b as the edge disc, versus the scan range.

6.4 What can go wrong with the slit scan alone: the beam sweep

It is clear which disc makes which edge: the front (early-TOF) edge of a peak is made by the disc that opens the slit (a), the back edge by the one that closes it (b). The difficulty lies elsewhere.

What "the beam sweep" is. A disc edge does not cut the beam instantly. The beam has a finite width, so the edge needs some time to cross it, and during that time the transmission changes gradually from 0 to 1 (Fig. 2). The model takes this time to be 367 μs, from the 40 mm beam width and 289 mm radius quoted in `hexaSub.c`. Neither is known precisely at the chopper positions, and the beam need not be uniform across its width. The true ramp could be 250 μs long, or have a different shape. That is what we mean by "the model's beam sweep may be wrong".

Why that matters. The peak a slit produces is roughly the slit width plus a ramp on each side, further smeared by the spread of departure times. A narrow slit with slow edges and a wide slit with fast edges produce almost the same peak (Fig. 8, left and middle).

- If the model's ramp is longer than the real one, the fit reproduces the observed peak by making the slit narrower: in our test 110 μs narrower for 367 instead of 250 μs.
- The slit width is (time b closes) − (time a opens). So a width error of 110 μs appears as an error of +55 μs for disc a and −55 μs for disc b, in opposite directions.
- The *centre* of the slit is not affected, because both sides move symmetrically.

The way out. What does not depend on the edges is the number of neutrons that get through. As long as the slit is wider than the ramp, its area is exactly its width: what is lost while the opening edge crosses the beam is made up while the closing edge crosses it (Fig. 8, right; proof in

Appendix B.2). Counting neutrons for a few slit widths therefore measures the true width, and with it the difference $\delta_b - \delta_a$, independently of the ramp. The slit scan gives the centre of the pair; this count gives the difference; together they give each edge separately. Section 6.5 describes the measurement.

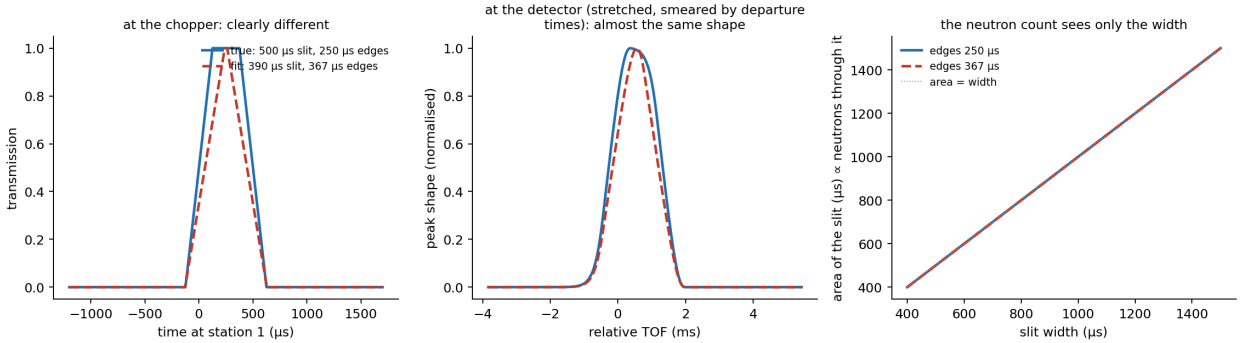


Figure 8: Left: a 500 μs slit with 250 μs edges (blue) and a 390 μs slit with 367 μs edges (red) at the chopper. Middle: the peaks they make at the detector after the stretch by L_{det}/L_1 and the smear by the departure-time spread, normalized; they are nearly indistinguishable, which is why a fit that assumes the wrong edges trades width for edge length. Right: the area of the slit, which sets the number of neutrons through it, equals the width whatever the edge length.

6.5 Separating disc a from disc b: the width scan

As Section 6.4 explains, the slit scan pins down the centre of each pair robustly, but its split between the opening disc (a) and the closing disc (b) rests on the assumed edge shape. Section 8.3 shows the size of the effect. The cure is a measurement that does not depend on the shape of the edges at all.

Area theorem. Let disc a open at o and disc b close at $c = o + w$. Each edge ramps over τ_b. If $w \geq \tau_b$, the area under the product of the two gates (the slit) is *exactly* w, whatever τ_b is and whatever the shape of the ramps (Appendix B.2). The departure-time spread only shifts neutrons in time and does not change the total either.

So the total number of neutrons through the slit is proportional to its true width $w_{\mathrm{true}} = w_{\mathrm{cmd}} - (\delta_b - \delta_a)$, times the flux per microsecond. We set five commanded widths, 800–1400 μs (all safely above τ_b even with a 300 μs offset error), and integrate the monitor counts of each run.

- **Simple version.** Draw a straight line through the five totals and extrapolate to zero counts. The zero is at $w_{\mathrm{cmd}} = \delta_b - \delta_a$.
- **Why it is not enough.** The flux per microsecond is not quite constant across a 1 ms slit, because the spectrum curves. In our simulation the straight-line extrapolation is off by 30–40 μs at stations 1–2.
- **What we do instead.** Fit the five totals with the instrument model, which includes the measured spectrum. Only one number is free per pair, $\delta_b - \delta_a$, plus a scale. This recovers the difference to better than 1 μs, *even when the model's beam sweep or emission shape is wrong.*

The runs need to be normalized to the same proton charge.

The slit scan supplies the centre of each pair, $(\delta_a + \delta_b)/2$; the width scan supplies the difference, $\delta_b - \delta_a$. Together they give both offsets.

6.6 The instrument model and the fit

Rather than locating peak centroids by hand, we fit all spectra at once with a forward model (`calibration_sim.py`, `chopper_model.py`):

- wavelengths on a fine grid (0.003 Å), weighted by the measured BL-6 spectrum of January 2025 (file `bl6_flux_2025A_Jan_rebinned.txt`);
- for each wavelength, departure times from the Gamma($k = 2$) distribution with mean $s \times$ `emission_delay`(λ), with s a free scale;
- every disc as a periodic gate with 367 μs ramps (Fig. 2), its centre at $P_i - \delta_i$;
- each surviving neutron deposited in the TOF histogram of the monitor and of the detector (Eq. 6, $d_0 = 0$ by convention). The deposit is shared between the two nearest 10 μs bins, so the model changes smoothly with every parameter.

The free parameters are the six δ_i and s, plus a free scale per spectrum (so detector efficiency and attenuation do not matter). They are found by Poisson-weighted least squares over the 36 slit-scan spectra (3 pairs $\times$ 6 wavelengths $\times$ 2 counters). The width scan then replaces the a/b split as described in Section 6.5.

7 One measurement, step by step

This section follows one run of the slit scan from the numbers typed into the control system to the line in the time–distance diagram. The run uses pair 1 as a 500 μs slit at 6 Å. The "instrument" is the simulated one of Section 8, whose real offsets differ from today's values by tens of microseconds.

Step 1: compute the six TotalDelay values from today's offsets

We do not need to know the true offsets to make a slit; the offsets in use today are good enough. Their error is probably below 100 μs, and the method measures exactly that error. For the slit we want:

- disc 1a opens 250 μs before, and disc 1b closes 250 μs after, the time at which 6 Å neutrons pass station 1 (the 6 Å flight time plus the drtsans mean delay);
- stations 2 and 3 stay wide open around the same wavelength, so that they do not cut into the bundle.

For each disc the wanted gap-centre time is converted with today's offset: TotalDelay = gap centre + offset, wrapped into one period. The four discs of stations 2 and 3 get exactly the values that an ordinary band request (here for a minimum wavelength of 4.19 Å) would give them; only the two discs of the pair under test differ. All six numbers are entered directly as the phases of the choppers. Table 3 shows the values; the planning tool prints such a table for every run of the scan (Section 9).

Table 3: The six TotalDelay values typed in for the example run (pair 1 as a 500 μs slit at 6 Å), computed from the offsets in use today (the IOC values). *Gap centre* is when each cut-out should cross the beam; TotalDelay = gap centre + current offset, modulo 16 667 μs.

Disc (EPICS)	role in this run	gap centre (μs)	current offset (μs)	TotalDelay (μs)
1a (5)	opens the slit	11454.5	15072.89	9860.8
1b (1)	closes the slit 500 μs later	5954.5	14954.46	4242.3
2a (6)	wide open around 6 Å	11889.6	14834.04	10056.9
2b (2)	wide open around 6 Å	11913.3	14805.40	10052.0
3a (3)	wide open around 6 Å	14525.0	14726.06	12584.4
3b (4)	wide open around 6 Å	14540.2	14565.60	12439.1

If today's offsets are wrong by Δ, the real slit is somewhat shifted and its width is $500\,\mu\text{s} - (\delta_b - \delta_a)_{\text{error}}$ instead of 500 μs. That does not matter: the shift and the width change are what the fit measures. There is no need to search for the slit by watching the detector. It is only worth confirming in the live display that a peak is there. If none is seen, today's offsets are worse than expected; widening the slit to 800 μs will find it.

Step 2: one run gives both spectra

One run produces one event NeXus file (`EQSANS_<run>.nxs.h5`). We checked a 2026 file (run 176505); it already contains everything the analysis needs:

- the beam-monitor events (`entry/monitor1`) and the detector events (`entry/bank*_events`). Each event carries its TOF measured from the *same* T0 pulse, so the two spectra share one time axis;
- the chopper settings of the run: requested phases `PhaseRequest1--6`, measured phases `Phase1--6`, speeds `Speed1--6`;
- the detector position `detectorZ` and the proton charge.

So no peak has to be read by hand and no run log has to be kept. Given the IPTS and the run numbers, the analysis tool opens each file, sums and rebins the monitor events and the detector events into one TOF spectrum each (10 μs bins of the recorded TOF), and writes them with the six phases into one ASCII file per run (Section 10). Figure 9 shows the two spectra of this run.

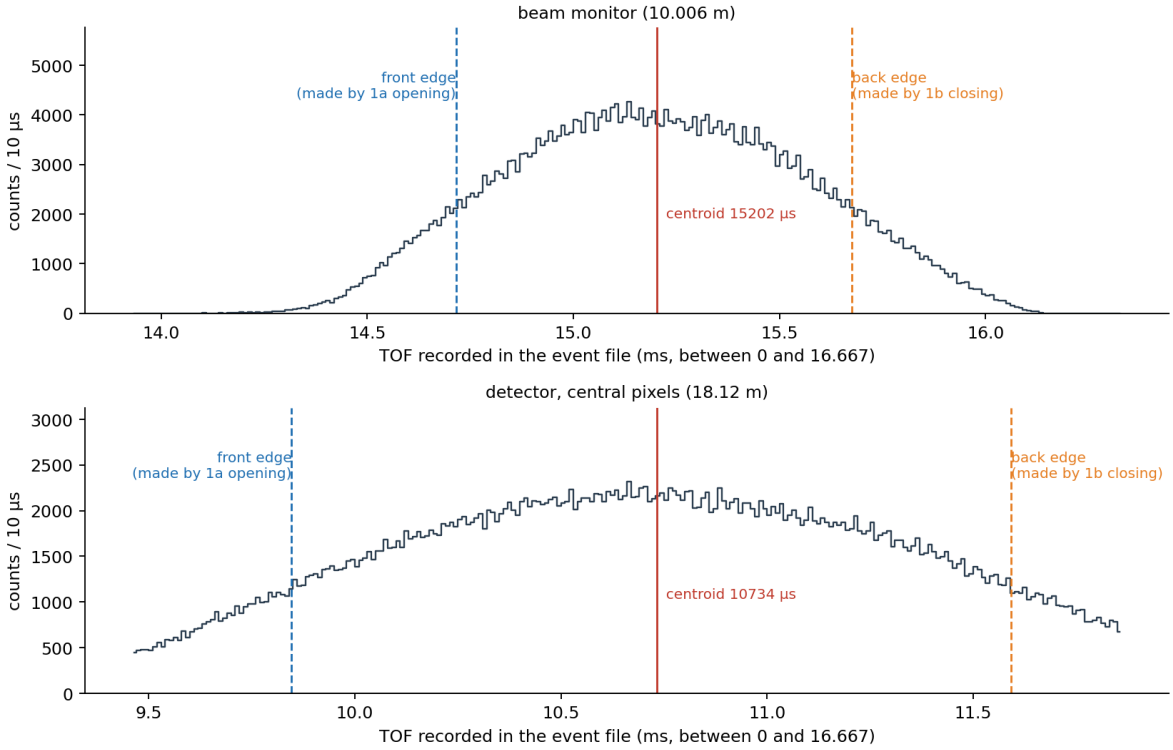


Figure 9: The two TOF spectra of the example run, both from the same event file. The recorded TOF runs from 0 to 16 667 μs (one source period). The **front edge** is the rising side of the peak (earlier TOF). It is made by the disc that opens the slit (1a), because the neutrons that arrive first are those that passed just as the slit opened. The **back edge** is the falling side, made by 1b closing. The detector peak is wider than the monitor peak: the detector is farther away, so the same spread in departure times and in slit time is stretched more.

Step 3: put both peaks on one time–distance diagram

Each peak is now a point in the time–distance diagram: horizontal coordinate = its TOF (time after T0); vertical coordinate = the distance of the counter from the moderator. Here that is (15 202 μs, 10.006 m) for the monitor and (10 734 μs, 18.12 m) for the detector. There is one subtlety. The DAS records the TOF modulo one source period of 16 667 μs. A 6 Å neutron needs about 27 500 μs to reach the detector, so its recorded 10 734 μs is really $10\,734 + 16\,667 = 27\,400\,\mu$s. Because we set the slit, we know the wavelength to within a few per cent, which is more than enough to choose the right number of periods to add ("unwrapping"). Figure 10 draws the result.

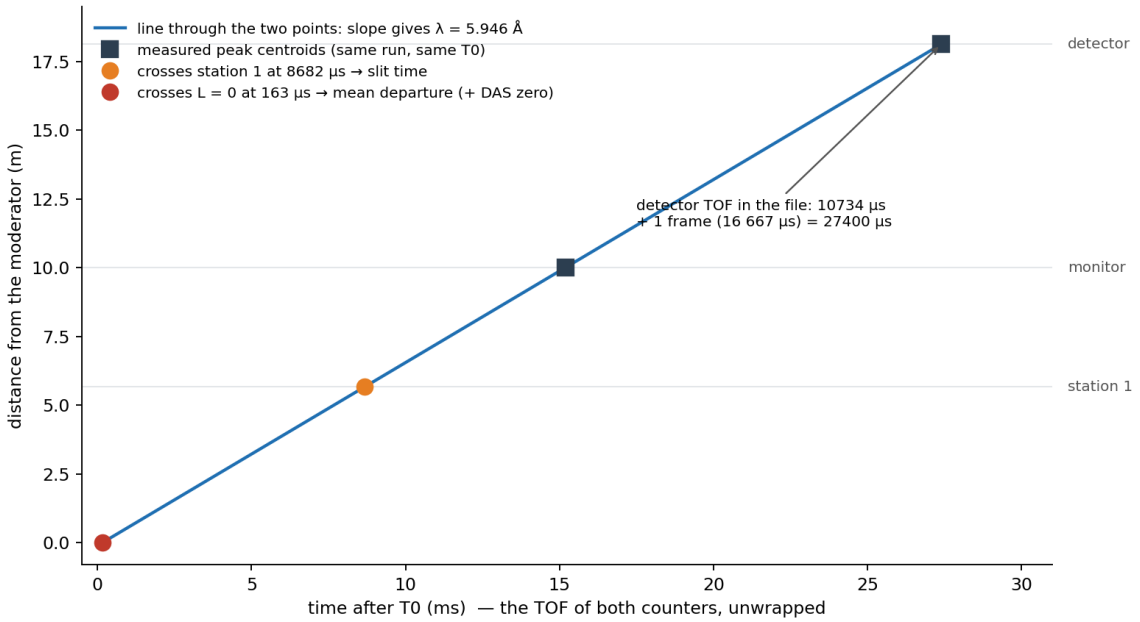


Figure 10: Both peaks of the example run in a time–distance diagram (real axes). The straight line through them is the path of the average neutron through the slit. Its slope gives the wavelength (5.95 Å). It crosses station 1 when the slit was open and $L = 0$ at the mean departure time plus the DAS zero.

Step 4: why a fit instead of the line

The two-point line is the idea, and it already gives a slit time within about 10–20 μs. A real peak is not a single neutron, though. It is a bundle with a skewed spread of departure times and a weighting by the spectrum, so the centroid of the peak is not exactly the path of one neutron of the average wavelength. (In the example the line meets $L = 0$ at 163 μs; the true mean departure plus DAS zero is about 153 μs.) The analysis therefore does not draw lines. It fits the instrument model of Section 6.6 to the full shapes of all 36 spectra at once. Each spectrum is shaped by the same six disc timings and the same moderator, so the shared parameters come out much more precisely than from any single peak. The line remains useful as a first guess and as a quick check during the experiment.

Step 5: the width scan

For the width scan nothing is fitted to peak shapes. For each run only the total number of monitor counts, divided by the proton charge, is used. Five runs per pair, with commanded widths of 800 to 1400 μs, give five numbers. The analysis finds the one value of $\delta_b - \delta_a$ for which the model reproduces them (Section 6.5).

8 Testing on a simulated instrument

8.1 Set-up

We built a “true” instrument that the methods do not know.

- Each disc’s real offset differs from Table 2 by a random amount between −150 and +150 μs.

- The real emission delay is 1.15 times the drtsans value.
- The DAS time zero is 6 μs late.

From this instrument we generated spectra with Poisson noise:

- the present method: one disc spinning, 10^6 monitor counts per disc;
- the slit scan: 4×10^5 counts per spectrum;
- the width scan: up to 10^6 monitor counts per run.

Each method then estimated the offsets from the spectra alone. Section 8.3 repeats this with the true emission shape and beam sweep also different from the model.

The quantity that matters for operation is *where a disc edge lands* when commanded to a target wavelength. We command edges at 3, 6 and 12 Å and compute, with the true instrument, the wavelength at which transmission crosses 50 %.

- With the present method's offsets, edges are commanded the way `calc6Phases` does it (departure at T0).
- With the proposed method's offsets, edges are commanded with the fitted emission delay. This is the "pulse-aware" phasing of the companion note on monochromatic mode.

8.2 Results

Table 4: Baseline (the model matches the true instrument). Top: error of the recovered offset, $\hat{\delta}_i - \delta_i$, mean ± s.d. over 3 trials (μs). Bottom: error of the delivered edge wavelength for an edge commanded at 6 Å (mean over trials, Å). Fitted emission scale: 1.083 ± 0.049 (true 1.15).

Offset error (μs)	1a	1b	2a	2b	3a	3b
present method	+81.9 ± 2.2	+94.1 ± 0.6	+44.4 ± 8.3	+42.1 ± 3.7	+12.3 ± 6.3	+11.6 ± 4.7
slit scan only	+6.0 ± 7.0	−1.9 ± 2.4	+0.7 ± 3.1	−1.1 ± 1.5	−5.0 ± 2.3	−1.5 ± 2.5
slit + width scan	+2.3 ± 4.2	+1.8 ± 2.2	−0.0 ± 3.2	−0.3 ± 1.4	−3.5 ± 1.8	−3.0 ± 0.8
Edge error at 6 Å (Å)	**1a**	**1b**	**2a**	**2b**	**3a**	**3b**
present method	−0.0416	−0.0329	−0.0494	−0.0505	−0.0538	−0.0540
slit scan only	−0.0011	−0.0065	−0.0035	−0.0044	−0.0052	−0.0037
slit + width scan	−0.0036	−0.0040	−0.0038	−0.0040	−0.0046	−0.0044

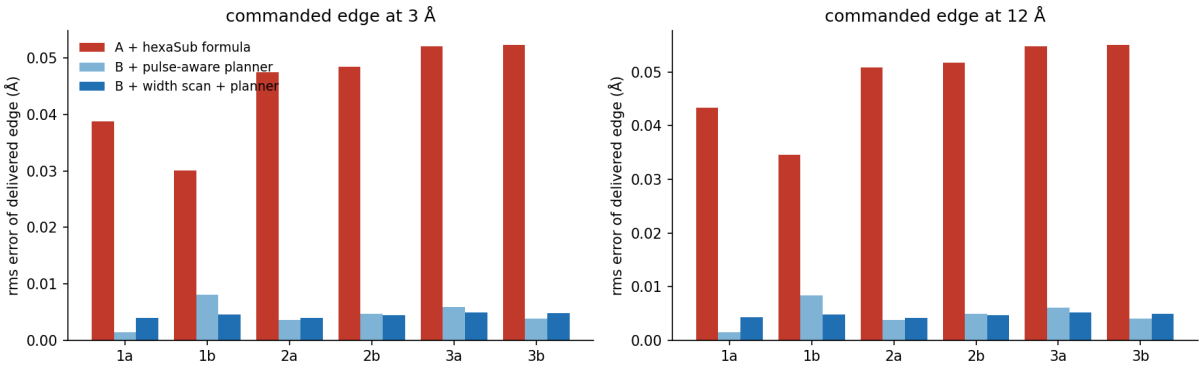


Figure 11: RMS error of the delivered edge wavelength per disc, for edges commanded at 3 Å and 12 Å. Red: present method with the present phase formula. Light blue: slit-scan fit alone. Dark blue: slit scan plus width scan.

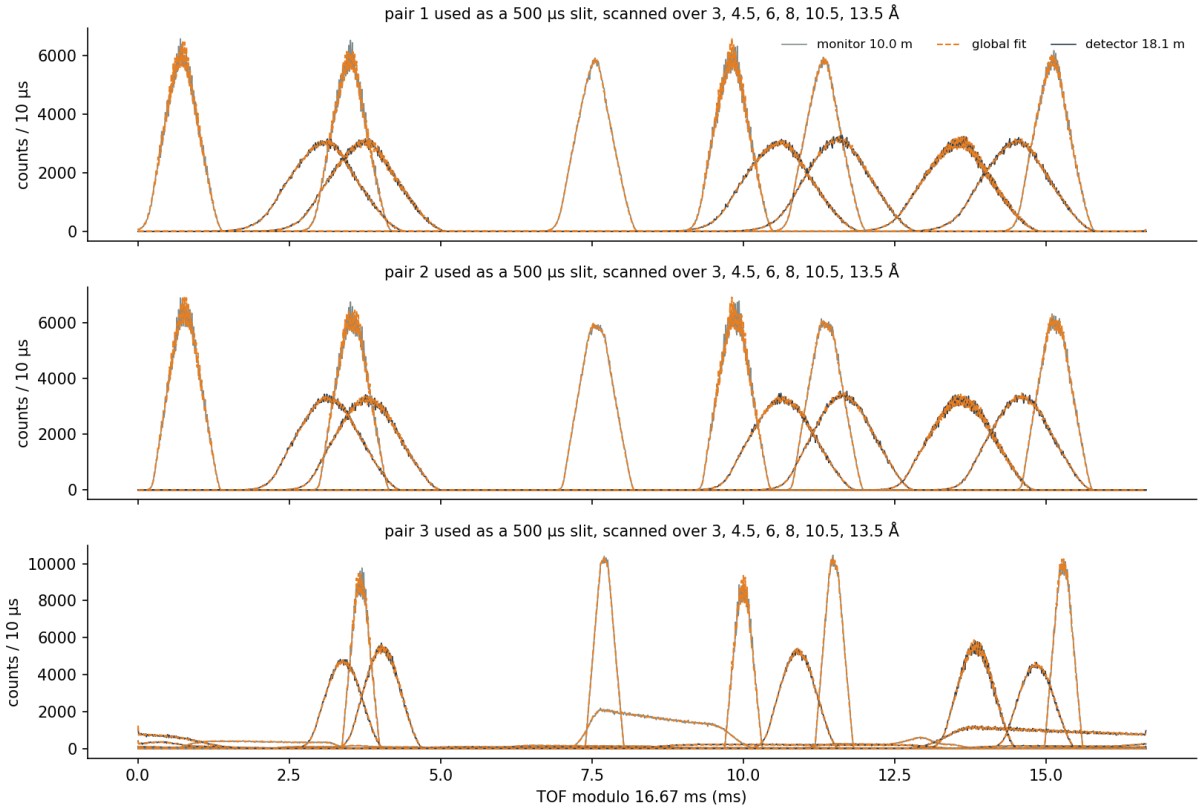


Figure 12: The 36 simulated slit-scan spectra of one trial (grey: monitor; black: detector) and the global fit (orange dashed). Each panel is one pair used as a 500 μs slit at six wavelengths. Detector peaks are wider than monitor peaks: the farther counter sees more of the departure-time spread, which is exactly what lets the fit separate departure time from disc time.

8.3 What if the model is wrong?

The fit assumes two things that we do not really know: the Gamma $k = 2$ shape of the departure times and the 367 μs beam sweep. We therefore built two more "true" instruments that break these

assumptions while the fit keeps them:

- departure times with an *exponential* distribution ($k = 1$), which has a much longer tail;
- a beam sweep of 250 μs instead of 367 μs, as for a narrower or non-uniform beam.

Table 5: Robustness. For each true instrument: the largest offset error over the six discs (absolute value of the trial mean, μs) and the largest RMS error of a delivered edge commanded at 6 Å (Å). The fit always assumes $k = 2$ and a 367 μs sweep.

	worst offset error (μs)			worst edge RMS at 6 Å (Å)		
True instrument	present	slit only	slit + width	present	slit only	slit + width
model = truth ($k = 2$, 367 μs)	94.1	6.0	3.5	0.0540	0.0082	0.0051
true emission exponential ($k = 1$)	72.0	29.2	8.2	0.0513	0.0250	0.0104
true beam sweep 250 μs	75.1	55.0	4.4	0.0532	0.0389	0.0029

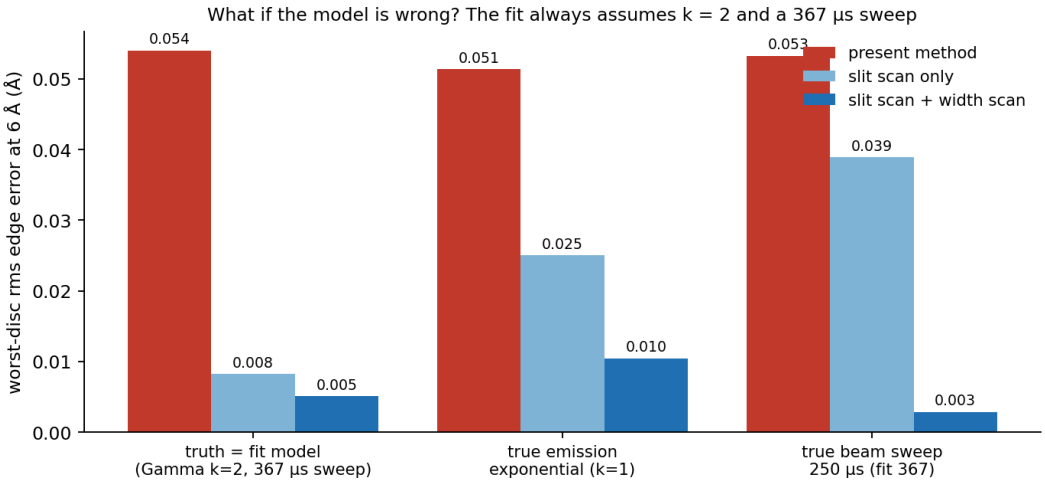


Figure 13: Worst-disc RMS edge error at 6 Å for each "true" instrument. The fit always assumes $k = 2$ and a 367 μs sweep.

8.4 Reading the results

- **The present method** returns offsets that are too large by 12–94 μs. The pattern follows Eq. 7: largest at station 1, smallest at station 3. The error-function bias adds some tens of microseconds on top. When these offsets are used with today's phase formula, every edge lands 0.03–0.05 Å short, almost the same for all discs, as Eq. 8 predicts. The result is the same for all three true instruments, because the bias comes from the method, not from the model.
- **The slit scan alone** recovers the offsets within a few microseconds when the model is right. It is not robust when the model is wrong. With a 250 μs true beam sweep it returns the opening and closing discs of each pair 50–60 μs off *in opposite directions*: the fit makes the slit narrower or wider to imitate steeper edges. The centre of each pair, the average of its two offsets, is still right to within a few microseconds in that case.

- **The slit scan plus width scan** fixes this. The width scan determines the difference between the two discs of a pair without using the edge shape. The worst offset error is 3.5 μs when the model is right, 4.4 μs with the wrong beam sweep and 8.2 μs with the wrong emission shape. The corresponding worst edge errors at 6 Å are 0.005, 0.003 and 0.010 Å.
- **The emission scale** fitted by the slit scan (1.08 $\pm$ 0.05 against a true 1.15) absorbs part of the DAS time zero. That is the free combination of Section 4.5, and as expected it does not spoil the edges.

9 How to do it on the instrument

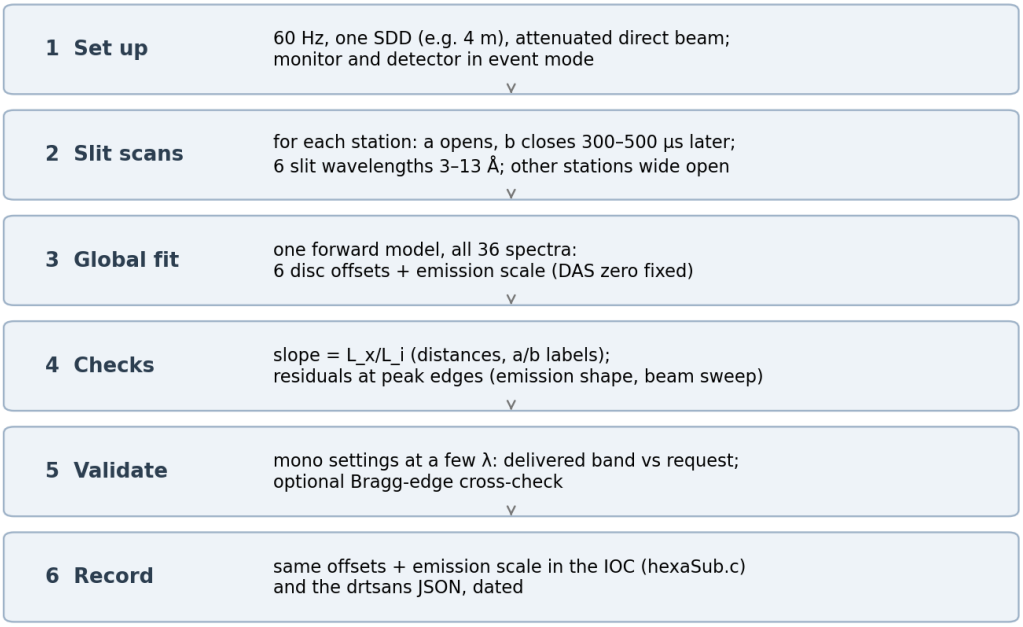


Figure 14: The procedure in six steps.

9.1 Do we need to know the current offsets?

For the analysis: no. The phases recorded in every run file (`Phase1--6`, `PhaseRequest1--6`) are the raw TotalDelay values, offset included. We checked this on run 176505: a phase delay of 1000 was entered and `Phase1` = 1000.0 is recorded. The fit therefore determines the offsets themselves, δ_i = recorded TotalDelay $-$ (time the gap centre actually crossed the beam). The result is the absolute value to put into `hexaSub.c` and drtsans. Current values serve only as a starting point for the fit and to recognize which pair formed the slit in each run. Values a few hundred microseconds wrong give the same answer.

For planning: only what the control system uses now, and the tool reads it. To tell you which numbers to type, the tool must know what the control system currently adds. It reads this from any recent run taken with an ordinary 60 Hz band request (a *reference run*). For each disc,

offset in use = recorded TotalDelay$-$gap-centre time that `calc6Phases` computes for that request.

The tool contains a port of `calc6Phases` for this. On a simulated reference run the six offsets came back within 0.25 μs. These offsets cancel in the end: the plan uses them to convert wanted times into TotalDelay values, and the analysis converts back with the same values.

9.2 Set-up

- Choppers at 60 Hz, SDD 4 m, the 4 m 2.5 Å transmission configuration (beamstop out), empty beam: nothing at the sample position.
- Monitor and detector in event mode, as usual.
- Beam slit 1 and proton charge per run as listed (Section 9.4): the attenuated d25_Cd for the 3 and 4.5 Å slits, the direct beam d5mm from 6 Å on. Within each width scan all five runs use the same slit.
- A reference run, to read the offsets in use: any recent run with an ordinary 60 Hz band request, all six discs spinning, at any SDD and wavelength.

9.3 What to type: the worksheet

If you are measuring now, the numbers are already in Appendix A: the full 33-step table, with beam slit and proton charge, made on 12 November 2026 from run 190564 (the most recent 60 Hz run at the time). Its offsets agree with the built-in drtsans values to 0.08 μs. Use it as it is unless the control system's offsets have been changed since. If they have, or to be sure, make a fresh table in one minute on analysis.sns.gov:

```
cd ~/slitcal   (or: git clone https://github.com/cw-do/slitcal.git)
PY=/usr/local/pixi/sans/.pixi/envs/default/bin/python
$PY slitcal.py plan --ipts <IPTS> --ref-run <recent 60 Hz run> --ref-ipts <its IPTS>
```

This writes `steps.txt` (instructions) and `worksheet.csv` (the same as a table, with an empty column for the run numbers) to `slitcal_IPTS-<ipts>/`. Each step lists the six TotalDelay values in beam order, 1a (Skf5), 1b (Skf1), 2a (Skf6), 2b (Skf2), 3a (Skf3), 3b (Skf4). For each step: set the six phases, take the run, write down the run number.

Two of the six values make the slit. The other four are those of an ordinary band request that contains the slit wavelength, so that the stations not under test let the slit's neutrons through. The request is not simply centred on the slit wavelength. When the station-3 pair forms a slit, its two 230° discs overlap a second time: each is open for 10.65 ms of the 16.67 ms period, so besides the slit they share a window of about $2 \times 10.65 - 16.67 - 0.5 \approx 4.1$ ms, some 6–10 ms after it. The band windows of stations 1 and 2 repeat every period too. With the band centred at 13.5 Å they also pass 3.0–3.7 Å neutrons from a later pulse, which go through that second window and outnumber the slit's own neutrons by about 200 to 1. The tool therefore tries every request whose band contains the slit wavelength and keeps the one that passes the largest share of neutrons within 1 Å of it: 87–98 % for station 3 (it was below 1 % at 13.5 Å), 100 % for stations 1 and 2, with the slit's own transmission unchanged. The two-disc slits of stations 1 and 2 (129.6° and 180°) have no second window.

9.4 The scan script

Instead of typing, the tool writes a script for the EQ-SANS scan server:

```
$PY slitcal.py script --ipts <IPTS for the runs> --ref-run 190564 --ref-ipts 37485
```

`slitcal_IPTS-<ipts>/eqsans_chopcal.py` uses the functions of `eqsans_scanfunctions_live`:

```
setipts(<ipts>)
loadconf('conf_4000mm_2p5A_60Hz_trans')    # 4 m, 60 Hz, beamslit1 d25_Cd
closeShutter()
# step  1: pair 1 slit at 3 A, 500 us
set_chopper_delay(5, 5572.07)    # 1a  <- slit
set_chopper_delay(1, 16620.30)   # 1b  <- slit
set_chopper_delay(6, 4628.85)    # 2a
...                              # 2b, 3a, 3b
openShutter()
runsampleid('T-chopcal 01/33 pair1 slit 3A 500us 4m d25_Cd', 0, 'peltier', 'pc', -1, 0.25)
closeShutter()
...
beamslit1('d5mm')                # from 6 A on: direct beam
...
set_choppers_auto()                        # Skf1-4
setpv('BL6:Chop:Skf5:OpModeSet', 1)        # Skf5 and Skf6
setpv('BL6:Chop:Skf6:OpModeSet', 1)
loadconf('conf_4000mm_2p5A_60Hz_trans')
```

`set_chopper_delay` puts a chopper in manual mode and sets its TotalDelay. The shutter stays closed while the phases change, because a half-set pair can let a white beam through. Position -1 leaves the sample table where it is. `set_choppers_auto()` returns only Skf1–4 to automatic mode, so the script does Skf5 and Skf6 with `setpv`. After the first run, check that its file records the six phases as set (`slitcal.py extract` prints them in the header).

Beam slit and proton charge. A slit run passes only a small part of the band, so the attenuated d25_Cd of the standard 2.5 Å transmission would make the long-wavelength runs very slow: 250 C (about 50 h) for the 33 runs. The direct beam d5mm is much brighter, but it must not overflow the detector. Both questions are answered from the 2026 flux measurements of IPTS-37618 (4 m, 60 Hz, six discs spinning): 25 runs with d25_Cd and d5mm, band and monochromatic settings from 2.5 to 14 Å. Each was modelled from its own recorded phases, and the ratio of measured to modelled counts gives a factor per slit:

- detector, d25_Cd: 0.032–0.037 from 2.5 to 14 Å. It is flat, so the model's wavelength spectrum is right;
- detector, d5mm: 0.47 at 2.5 Å rising to 2.4 at 10 Å, i.e. 15 to 70 times d25_Cd (fitted as $\propto \lambda^{1.16}$);
- monitor: the same for both slits, rising steeply with wavelength (its efficiency).

The largest detector peak already recorded with d5mm, without trouble, is 2.8×10^5 counts per C per 10 μs bin (run 186454: 2.5 Å, 10 % spread). A run uses d5mm when its predicted peak stays below 0.8 of that, otherwise d25_Cd. This gives d25_Cd for the 3 and 4.5 Å slits and d5mm for everything from 6 Å, with peaks at most 0.62 of the recorded one. A smaller slit is not needed. The proton charge is set for 4×10^5 counts in each counter (slit scan) and 10^6 monitor counts in the narrowest run of a width scan (one value for all five widths). Total: 9.3 C, about 2 h at the 2026 beam power, plus the time the choppers take to re-phase between runs.

9.5 Option: spin only the pair under test

With `--mode parked` only the two discs of the pair under test spin; the other four are parked open. Each step then has two values instead of six. This is *not required*: the default above works and keeps the other stations removing unwanted wavelengths. It is offered because it is simpler on the instrument, and it isolates the pair completely, so the result does not depend at all on the other discs.

The price is that nothing removes the slit's aliases. The slit at station 1 opens again every 16.67 ms, so it also passes wavelengths 11.6 Å, 23 Å, ... longer than intended, and slow neutrons from earlier pulses. They are weak, they arrive at other TOFs, and the model includes them (its wavelength range is extended to 35 Å for such runs). Both modes were tested end to end (Section 10). For station 3 the price is higher: with stations 1 and 2 parked, the second overlap window of the 230° discs (4.1 ms) passes every wavelength that reaches it, and these runs are dominated by neutrons that are not in the slit. Use the default mode for station 3 if at all possible.

9.6 Run list

Table 6: Runs for a full calibration. Counts are those used in the simulation; the statistical error of the offsets scales as $1/\sqrt{\text{counts}}$.

Block	Settings	Runs	Counts per run
Reference	any ordinary 60 Hz band request (can be an existing run)	0–1	–
Slit scan	each pair as a 500 μs slit centred at 3, 4.5, 6, 8, 10.5, 13.5 Å; the other two pairs set by a band request that contains it (or parked open)	18	$\geq 4 \times 10^5$ each counter
Width scan	each pair centred at 6 Å, widths 800, 950, 1100, 1250, 1400 μs	15	$\sim 10^6$ monitor
Validation	monochromatic settings at, e.g., 4, 8, 12 Å with 5 % bands	3	–

9.7 Analysis

All of the following is done by the tool of Section 10. Give it the IPTS and the run numbers and press *Fit* (or run `slitcal.py analyze --ipts 37424 --runs 180100-180132`). It starts from the offsets the plan was made with (`offsets_in_use.json` in the same folder).

1. Histogram monitor and detector TOF (10 μs bins, one frame) and read the six phases of each run.
2. Run the global fit (Section 6.6) on the 36 slit-scan spectra: six offsets and the emission scale.
3. Run the width-scan fit for each pair and replace the a/b split.
4. **Checks.**
   - The peak position must move at L_x/L_i times the slit time (Fig. 7); a mismatch means a wrong distance or swapped a/b labels.

- Residuals should be featureless. Systematic residuals at the peak flanks point to the emission shape or the beam sweep. Both can be freed in the fit if needed.

5. **Validation.** Set a few monochromatic bands and confirm that the delivered band (from drtsans or from the monitor TOF) matches the request. Optionally confirm the absolute wavelength with a Bragg edge, for example a beryllium filter (3.96 Å). The method does not need it, but it is an independent check.

6. **Record.** Put the same offsets in the IOC (`hexaSub.c`) and in the drtsans configuration file, with the date. Put the fitted emission scale in both too, so that the control system and the reduction use the same departure-time model.

9.8 Keeping it calibrated

After any change to a disc or controller, and perhaps once per cycle, repeat one slit setting per pair at one wavelength. A shift of the monitor and detector peaks then shows directly whether an offset has moved.

10 The analysis tool: `slitcal`

`slitcal` does the planning and the analysis. It is meant to be cloned with git onto analysis.sns.gov and run there, next to the data. It needs only numpy, scipy, h5py and matplotlib, all present in the `sans` environment (`/usr/local/pixi/sans/.pixi/envs/default/bin/python`). There is a command-line program, `slitcal.py`, and a window, `slitcal_gui.py`; both call the same functions.

The input is an IPTS number and a list of run numbers; the tool finds the files itself as `/SNS/EQSANS/IPTS-<ipts>/nexus/EQSANS_<run>.nxs.h5`. It works in two stages, so that the large event files are read only once.

1. **Extract.** Every run's monitor and detector events are summed and rebinned into one TOF spectrum each (10 μs bins, one frame). Both are written into one ASCII file per run, `slitcal_IPTS-<ipts>/spectra/EQSANS_<run>_spectra.dat`: three columns (TOF bin centre, monitor counts, detector counts). The header lines carry the six phases, the six speeds, the detector distance and the proton charge. Runs already extracted are not read again.

2. **Fit.** The analysis reads only these ASCII files. They can be inspected, plotted or re-fitted with any other program.

What the user does

1. **Plan.** `python slitcal.py plan --ipts 37424 --ref-run 180099`, or tab 1 of the window (IPTS, reference run, *Read reference run*). The tool reads the offsets in use from the reference run and prints the 33 steps with the six TotalDelay values each (Fig. 15; `--mode parked` for the two-disc variant).

2. **Measure.** For each step, enter the six values as the chopper phases, take the run and write down the run number. Even that is only a convenience: the tool recognizes every run from the phases stored in its own file, so only the range of run numbers is needed.

3. **Analyse.** `python slitcal.py analyze --ipts 37424 --runs 180100-180132`, or tab 2 of the window (IPTS, runs, *Extract runs → ASCII and load*, then *Fit*). For every run the tool:

- sums and rebins the monitor and detector events and writes the ASCII file;
- reads `Phase1--6` (falling back to `PhaseRequest1--6`), `Speed1--6`, `detectorZ` and the proton charge;
- finds the pair whose two discs form a short slit, and from the phases its wavelength and width;
- sorts the run into the slit scan (width below 700 μs) or the width scan.

Discs at speed zero count as parked open. Runs in which no spinning pair forms a slit, or in which a disc spins at another speed, are skipped and listed in the log (Fig. 16).

4. **Result.** Tab 3 (Fig. 17) and the output folder hold:
   - a table of reference offset, correction and new offset for each disc, at 60 and at 30 Hz;
   - the fitted emission scale;
   - figures of every spectrum with the fit, the width scan and the corrections.

   `offsets.csv` and `results.json` hold the same numbers for the IOC and the drtsans file.

Test on synthetic data

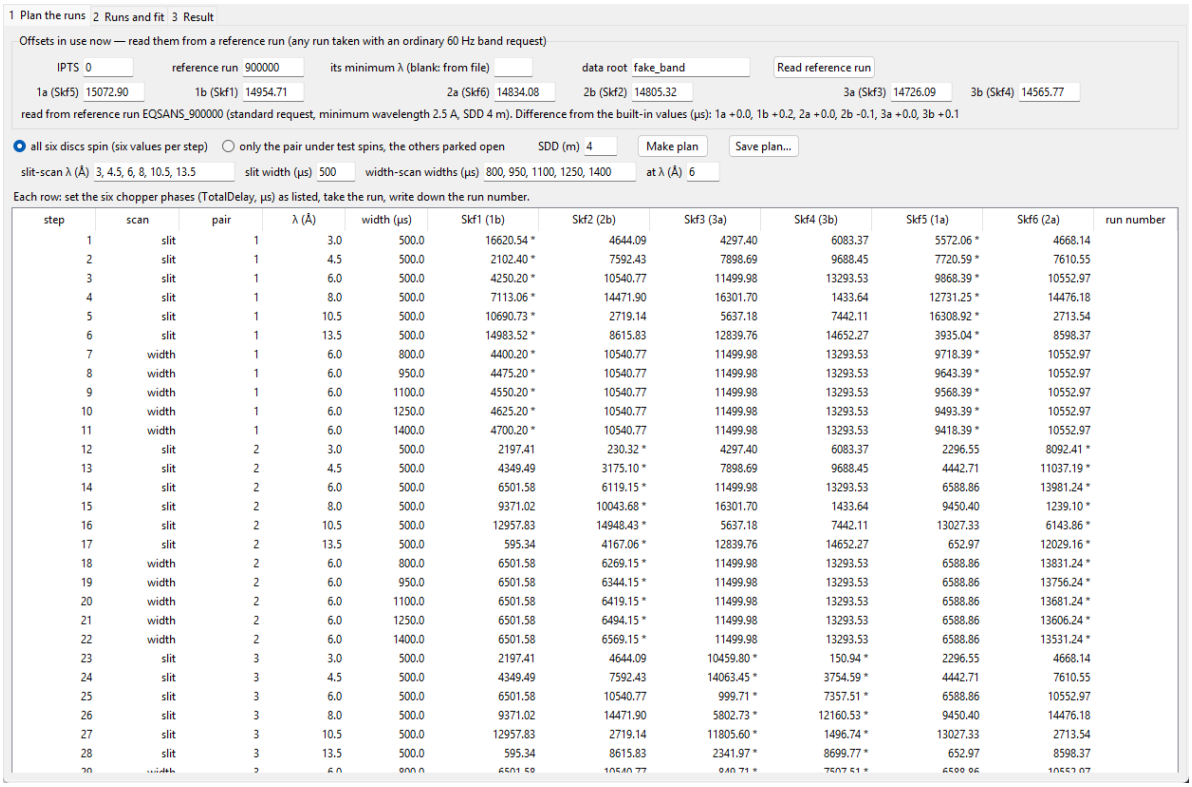
Real calibration runs do not exist yet, so the tool was tested end to end on synthetic data. `slitcal.py simulate plan.csv` writes event NeXus files with the same layout as real EQ-SANS files: `entry/monitor1`, `entry/bank*_events`, and the `DASlogs` entries above. It writes them into a directory tree that mimics the archive (`<root>/IPTS-0/nexus/EQSANS_9000xx.nxs.h5`). The files come from a hidden instrument whose truth is saved separately. The control system of this instrument uses the built-in offsets, while the real offsets differ from them by up to 150 μs. Run 900000 is a reference run with an ordinary 2.5 Å request. The test was done exactly as on the cluster, using only the files:

1. `plan --ipts 0 --ref-run 900000` recovered the offsets in use from the reference run to within 0.25 μs;
2. the 33 steps were simulated, for both the six-disc and the parked-open variant;
3. `analyze --data-root <root> --ipts 0 --runs 900001-900033` found the true offsets.

Table 7 shows the result: all six offsets within 7 μs with six discs spinning and within 9 μs with the other discs parked. The fit's χ^2 per degree of freedom was 3 to 9 on these very high-statistics spectra. Part of the true instrument (its DAS zero) cannot be represented exactly by a fit that fixes it to zero, and that small mismatch shows at 4×10^5 counts. It does not affect the offsets. On real data χ^2 should be read as a relative diagnostic, comparing runs and pairs, not as a pass/fail number.

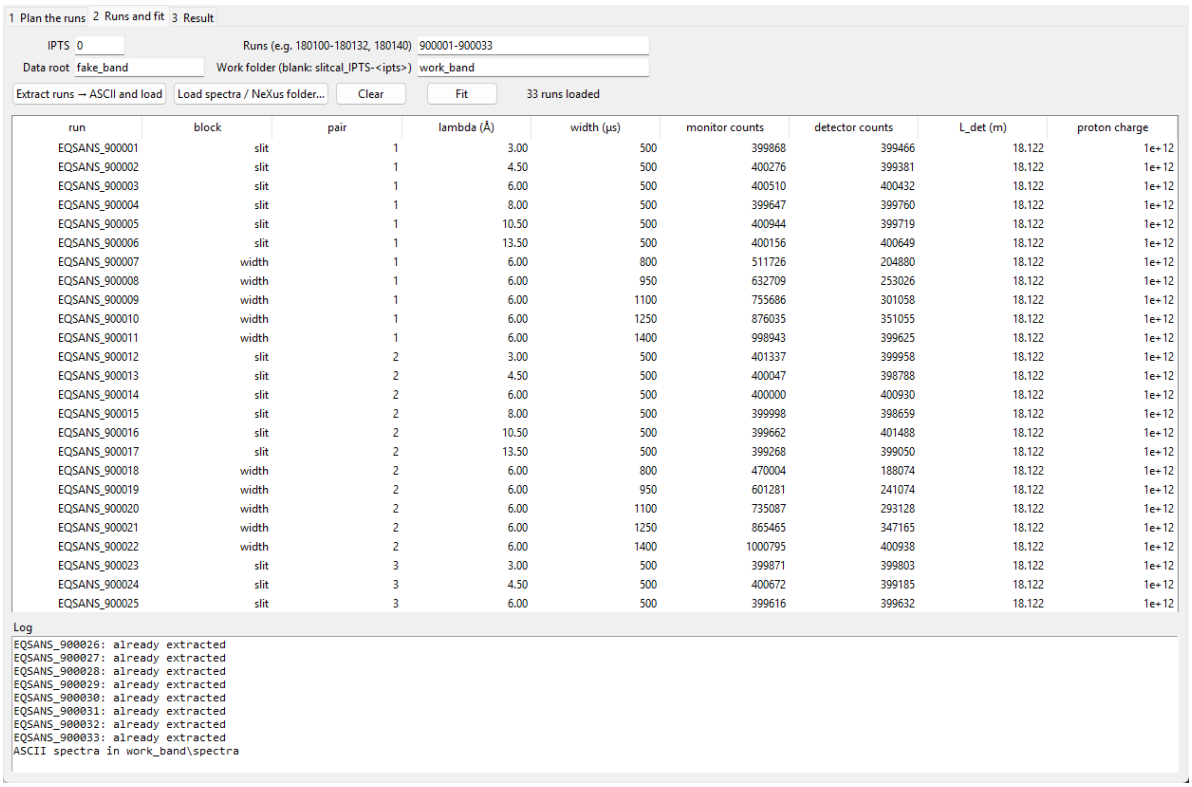
Table 7: End-to-end test of `slitcal`: a reference run and 33 synthetic event NeXus files per variant, written by `slitcal.py simulate` from a hidden instrument (control system using the built-in offsets; true offsets differ by up to 150 μs; emission 1.15× drtsans; DAS zero +6 μs), planned from the reference run and analysed from the files alone.

			six discs spinning		pair only, others parked	
Disc (Skf)	in use (μs)	true (μs)	recovered	error	recovered	error
1a (5)	15072.89	15076.44	15083.43	+7.0	15073.99	−2.4
1b (1)	14954.46	15089.60	15095.90	+6.3	15085.10	−4.5
2a (6)	14834.04	14727.29	14731.58	+4.3	14721.87	−5.4
2b (2)	14805.40	14939.99	14942.52	+2.5	14937.03	−3.0
3a (3)	14726.06	14669.61	14668.53	−1.1	14668.02	−1.6
3b (4)	14565.69	14542.69	14541.68	−1.0	14534.00	−8.7



step	scan	pair	λ (Å)	width (µs)	Skf1 (1b)	Skf2 (2b)	Skf3 (3a)	Skf4 (3b)	Skf5 (1a)	Skf6 (2a)	run number
1	slit	1	3.0	500.0	16620.54 *	4644.09	4297.40	6083.37	5572.06 *	4668.14	
2	slit	1	4.5	500.0	2102.40 *	7592.43	7898.69	9688.45	7720.59 *	7610.55	
3	slit	1	6.0	500.0	4250.20 *	10540.77	11499.98	13293.53	9868.39 *	10552.97	
4	slit	1	8.0	500.0	7113.06 *	14471.90	16301.70	1433.64	12731.25 *	14476.18	
5	slit	1	10.5	500.0	10690.73 *	2719.14	5637.18	7442.11	16308.92 *	2713.54	
6	slit	1	13.5	500.0	14983.52 *	8615.83	12839.76	14652.27	3935.04 *	8598.37	
7	width	1	6.0	800.0	4400.20 *	10540.77	11499.98	13293.53	9718.39 *	10552.97	
8	width	1	6.0	950.0	4475.20 *	10540.77	11499.98	13293.53	9643.39 *	10552.97	
9	width	1	6.0	1100.0	4550.20 *	10540.77	11499.98	13293.53	9568.39 *	10552.97	
10	width	1	6.0	1250.0	4625.20 *	10540.77	11499.98	13293.53	9493.39 *	10552.97	
11	width	1	6.0	1400.0	4700.20 *	10540.77	11499.98	13293.53	9418.39 *	10552.97	
12	slit	2	3.0	500.0	2197.41	230.32 *	4297.40	6083.37	2296.55	8092.41 *	
13	slit	2	4.5	500.0	4349.49	3175.10 *	7898.69	9688.45	4442.71	11037.19 *	
14	slit	2	6.0	500.0	6501.58	6119.15 *	11499.98	13293.53	6588.86	13981.24 *	
15	slit	2	8.0	500.0	9371.02	10043.68 *	16301.70	1433.64	9450.40	1239.10 *	
16	slit	2	10.5	500.0	12957.83	14948.43 *	5637.18	7442.11	13027.33	6143.86 *	
17	slit	2	13.5	500.0	595.34	4167.06 *	12839.76	14652.27	652.97	12029.16 *	
18	width	2	6.0	800.0	6501.58	6269.15 *	11499.98	13293.53	6588.86	13831.24 *	
19	width	2	6.0	950.0	6501.58	6344.15 *	11499.98	13293.53	6588.86	13756.24 *	
20	width	2	6.0	1100.0	6501.58	6419.15 *	11499.98	13293.53	6588.86	13681.24 *	
21	width	2	6.0	1250.0	6501.58	6494.15 *	11499.98	13293.53	6588.86	13606.24 *	
22	width	2	6.0	1400.0	6501.58	6569.15 *	11499.98	13293.53	6588.86	13531.24 *	
23	slit	3	3.0	500.0	2197.41	4644.09	10459.80 *	150.94 *	2296.55	4668.14	
24	slit	3	4.5	500.0	4349.49	7592.43	14063.45 *	3754.59 *	4442.71	7610.55	
25	slit	3	6.0	500.0	6501.58	10540.77	999.71 *	7357.51 *	6588.86	10552.97	
26	slit	3	8.0	500.0	9371.02	14471.90	5802.73 *	12160.53 *	9450.40	14476.18	
27	slit	3	10.5	500.0	12957.83	2719.14	11805.60 *	1496.74 *	13027.33	2713.54	
28	slit	3	13.5	500.0	595.34	8615.83	2341.97 *	8699.77 *	652.97	8598.37	

Figure 15: `slitcal_gui.py`, tab 1: the offsets in use read from a reference run, and the worksheet: six TotalDelay values per step, the two that make the slit marked with an asterisk.



run	block	pair	lambda (Å)	width (µs)	monitor counts	detector counts	L_det (m)	proton charge
EQSANS_900001	slit	1	3.00	500	399868	399466	18.122	1e+12
EQSANS_900002	slit	1	4.50	500	400276	399381	18.122	1e+12
EQSANS_900003	slit	1	6.00	500	400510	400432	18.122	1e+12
EQSANS_900004	slit	1	8.00	500	399647	399760	18.122	1e+12
EQSANS_900005	slit	1	10.50	500	400944	399719	18.122	1e+12
EQSANS_900006	slit	1	13.50	500	400156	400649	18.122	1e+12
EQSANS_900007	width	1	6.00	800	511726	204880	18.122	1e+12
EQSANS_900008	width	1	6.00	950	632709	253026	18.122	1e+12
EQSANS_900009	width	1	6.00	1100	755686	301058	18.122	1e+12
EQSANS_900010	width	1	6.00	1250	876035	351055	18.122	1e+12
EQSANS_900011	width	1	6.00	1400	998943	399625	18.122	1e+12
EQSANS_900012	slit	2	3.00	500	401337	399958	18.122	1e+12
EQSANS_900013	slit	2	4.50	500	400047	398788	18.122	1e+12
EQSANS_900014	slit	2	6.00	500	400000	400930	18.122	1e+12
EQSANS_900015	slit	2	8.00	500	399998	398659	18.122	1e+12
EQSANS_900016	slit	2	10.50	500	399662	401488	18.122	1e+12
EQSANS_900017	slit	2	13.50	500	399268	399050	18.122	1e+12
EQSANS_900018	width	2	6.00	800	470004	188074	18.122	1e+12
EQSANS_900019	width	2	6.00	950	601281	241074	18.122	1e+12
EQSANS_900020	width	2	6.00	1100	735087	293128	18.122	1e+12
EQSANS_900021	width	2	6.00	1250	865465	347165	18.122	1e+12
EQSANS_900022	width	2	6.00	1400	1000795	400938	18.122	1e+12
EQSANS_900023	slit	3	3.00	500	399871	399803	18.122	1e+12
EQSANS_900024	slit	3	4.50	500	400672	399185	18.122	1e+12
EQSANS_900025	slit	3	6.00	500	399616	399632	18.122	1e+12

Figure 16: Tab 2: IPTS and run numbers in; the 33 synthetic runs extracted to ASCII and classified from the phases in each file.

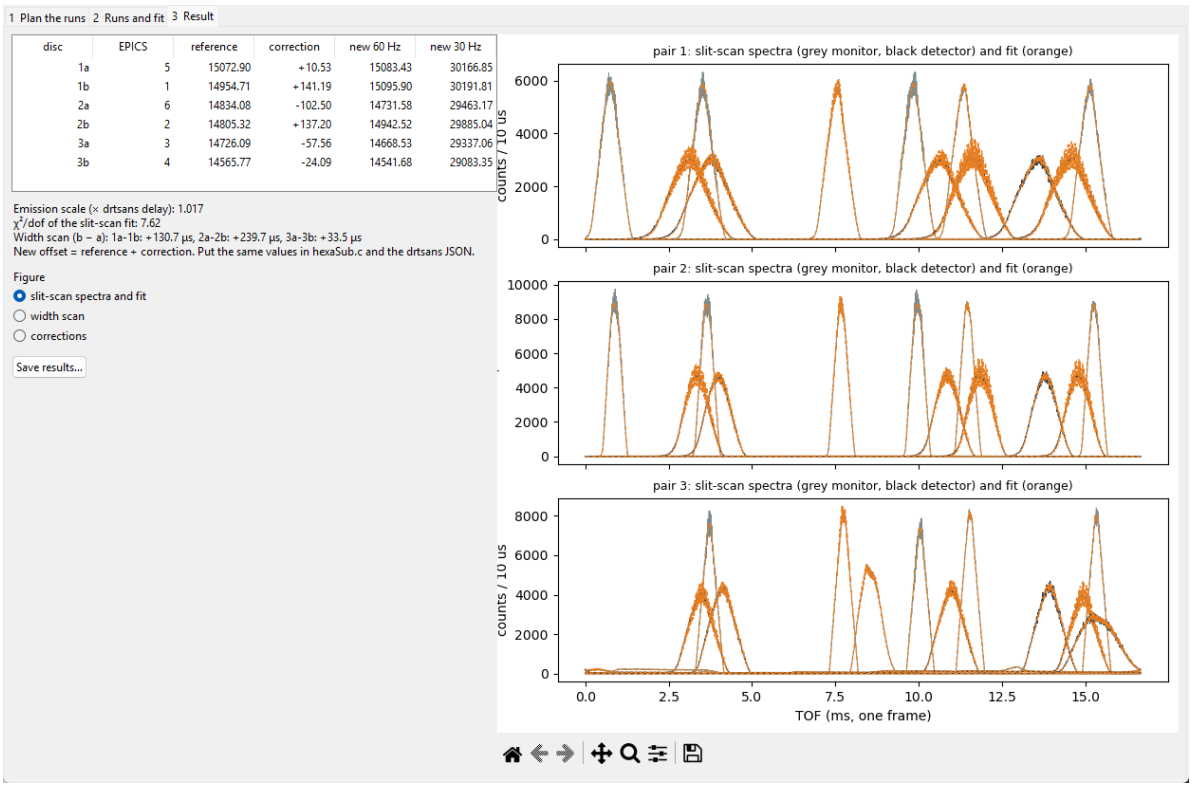


Figure 17: Tab 3: new offsets and the fit of the slit-scan spectra.

What the tool does not do yet

- It uses all detector pixels. With the attenuated direct beam this is dominated by the central pixels, but a pixel mask for the beam spot would be cleaner.
- The moderator-to-sample distance is fixed at 14.122 m, the value used by `hexaSub.c`; the detector distance is taken from `detectorZ`.
- It has not yet seen real data, so its first use should be checked against the slope test and the residual plots before the new offsets are installed.

11 Limitations and open points

- **Simulation only.** Everything here is simulated. The model's weakest parts are the departure-time shape and the beam profile behind the 367 μs ramp. The sensitivity tests show the proposed method tolerates large errors in both. Real data may still show effects not in the model, for example a wavelength-dependent monitor efficiency (it can be added as a $1/v$ term) or chopper phase jitter, which widens edges but does not move them on average.
- **Normalization.** The width scan assumes the runs are normalized to proton charge to well below 1 %. A 0.5 % normalization error moves the a/b split by a few microseconds.
- **Existing data.** The existing calibration runs (`edge_refinement`, IPTS-37424) use one disc and one counter, so they cannot be re-analysed with this method. They could be refitted with

the forward model to remove the error-function bias, but they would still need an absolute anchor for the departure time.

- **Real data.** `slitcal` reads real event files, and it has planned from a real reference run (run 190564, Appendix A). The fit has so far been run only on synthetic files (Section 10).

A Worksheet for the instrument as of 12 November 2026

These tables were written on analysis.sns.gov by `slitcal.py script --ref-run 190564 --ref-ipts 37485` and, for the second table, `slitcal.py plan ... --mode parked` (slitcal commit 76bb4fe). The same command writes the scan script of Section 9.4 with exactly these values. The reference run 190564 was taken at 60 Hz, minimum wavelength 15 Å, SDD 9 m. The offsets the control system was using, read from it, are (μs): 1a (Skf5) 15072.90, 1b (Skf1) 14954.47, 2a (Skf6) 14834.03, 2b (Skf2) 14805.38, 3a (Skf3) 14726.04, 3b (Skf4) 14565.61. They agree with the built-in drtsans values within 0.08 μs.

How to use the table. All values are TotalDelay in microseconds, the number typed into the phase field of each chopper. The columns are in beam order, from upstream to downstream, and each column header gives the disc position and its EPICS chopper number:

Position	1a	1b	2a	2b	3a	3b
EPICS chopper	Skf5	Skf1	Skf6	Skf2	Skf3	Skf4
Distance from moderator (m)	5.660	5.676	7.760	7.776	9.498	9.508

For every row:

1. set beam slit 1 and the six phases as listed (choppers at 60 Hz, 4 m 2.5 Å transmission configuration, empty beam), with the shutter closed;
2. open the shutter and take the run to the listed proton charge; close the shutter;
3. write the run number in the last column.

Bold values are the two discs that make the slit. The other values are those of a band request containing the slit wavelength, chosen so that nothing else leaks through (Section 9). The last columns give beam slit 1 and the proton charge for the run (Section 9.4). The order of the rows does not matter, and a repeated or extra run does no harm: the analysis recognises each run from the phases stored in its own file. At the end, give the tool the IPTS and the run-number range.

These numbers are valid as long as the offsets in the control system are unchanged. If they have been changed, make a new table (Section 9).

A.1 Default: all six discs spinning

Step	Scan	Pair	λ (Å)	Width (μs)	**1a** Skf5	**1b** Skf1	**2a** Skf6	**2b** Skf2	**3a** Skf3	**3b** Skf4	Beam slit 1	PC (C)	Run no.
1	slit	1	3	500	**5572.07**	**16620.30**	4628.85	4604.83	4249.33	6035.15	d25_Cd	0.25	
2	slit	1	4.5	500	**7720.59**	**2102.16**	7571.27	7553.18	7850.62	9640.23	d25_Cd	0.55	
3	slit	1	6	500	**9868.39**	**4249.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
4	slit	1	8	500	**12731.25**	**7112.81**	14436.90	14432.65	16253.63	1385.42	d5mm	0.10	
5	slit	1	10.5	500	**16308.92**	**10690.48**	2674.25	2679.89	5589.11	7393.88	d5mm	0.25	

Step	Scan	Pair	λ (Å)	Width (μs)	**1a** Skf5	**1b** Skf1	**2a** Skf6	**2b** Skf2	**3a** Skf3	**3b** Skf4	Beam slit 1	PC (C)	Run no.
6	slit	1	13.5	500	**3935.04**	**14983.28**	8559.08	8576.57	12791.69	14604.04	d5mm	0.65	
7	width	1	6	800	**9718.39**	**4399.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
8	width	1	6	950	**9643.39**	**4474.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
9	width	1	6	1100	**9568.39**	**4549.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
10	width	1	6	1250	**9493.39**	**4624.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
11	width	1	6	1400	**9418.39**	**4699.96**	10513.68	10501.52	11451.91	13245.31	d5mm	0.10	
12	slit	2	3	500	2267.94	2168.47	**8092.36**	**230.38**	4249.33	6035.15	d25_Cd	0.35	
13	slit	2	4.5	500	4414.09	4320.55	**11037.14**	**3175.16**	7850.62	9640.23	d25_Cd	0.75	
14	slit	2	6	500	6560.25	6472.64	**13981.19**	**6119.21**	11451.91	13245.31	d5mm	0.10	
15	slit	2	8	500	9421.79	9342.09	**1239.05**	**10043.73**	16253.63	1385.42	d5mm	0.15	
16	slit	2	10.5	500	12998.71	12928.90	**6143.81**	**14948.49**	5589.11	7393.88	d5mm	0.35	
17	slit	2	13.5	500	624.35	566.40	**12029.10**	**4167.12**	12791.69	14604.04	d5mm	0.90	
18	width	2	6	800	6560.25	6472.64	**13831.19**	**6269.21**	11451.91	13245.31	d5mm	0.15	
19	width	2	6	950	6560.25	6472.64	**13756.19**	**6344.21**	11451.91	13245.31	d5mm	0.15	
20	width	2	6	1100	6560.25	6472.64	**13681.19**	**6419.21**	11451.91	13245.31	d5mm	0.15	
21	width	2	6	1250	6560.25	6472.64	**13606.19**	**6494.21**	11451.91	13245.31	d5mm	0.15	
22	width	2	6	1400	6560.25	6472.64	**13531.19**	**6569.21**	11451.91	13245.31	d5mm	0.15	
23	slit	3	3	500	1409.48	1307.63	3451.89	3425.49	**10459.75**	**150.78**	d25_Cd	0.45	
24	slit	3	4.5	500	3555.63	3459.72	6394.30	6373.84	**14063.40**	**3754.43**	d25_Cd	0.90	
25	slit	3	6	500	5701.79	5611.81	9336.72	9322.18	**999.65**	**7357.35**	d5mm	0.10	
26	slit	3	8	500	8563.33	8481.25	13259.93	13253.31	**5802.67**	**12160.37**	d5mm	0.15	
27	slit	3	10.5	500	12140.25	12068.06	1497.29	1500.55	**11805.55**	**1496.58**	d5mm	0.35	
28	slit	3	13.5	500	2055.12	2001.12	10520.69	10542.14	**2341.92**	**8699.62**	d5mm	0.90	
29	width	3	6	800	5558.71	5468.33	9140.55	9125.63	**849.65**	**7507.35**	d5mm	0.15	
30	width	3	6	950	5558.71	5468.33	9140.55	9125.63	**774.65**	**7582.35**	d5mm	0.15	
31	width	3	6	1100	5558.71	5468.33	9140.55	9125.63	**699.65**	**7657.35**	d5mm	0.15	
32	width	3	6	1250	5558.71	5468.33	9140.55	9125.63	**624.65**	**7732.35**	d5mm	0.15	
33	width	3	6	1400	5558.71	5468.33	9140.55	9125.63	**549.65**	**7807.35**	d5mm	0.15	

A.2 Option: only the pair under test spins

The four discs marked *park* are stopped in the open position. Only the two listed discs spin at 60 Hz.

Step	Scan	Pair	λ (Å)	Width (μs)	**1a** Skf5	**1b** Skf1	**2a** Skf6	**2b** Skf2	**3a** Skf3	**3b** Skf4	Run no.
1	slit	1	3	500	**5572.07**	**16620.30**	*park*	*park*	*park*	*park*	
2	slit	1	4.5	500	**7720.59**	**2102.16**	*park*	*park*	*park*	*park*	
3	slit	1	6	500	**9868.39**	**4249.96**	*park*	*park*	*park*	*park*	
4	slit	1	8	500	**12731.25**	**7112.81**	*park*	*park*	*park*	*park*	
5	slit	1	10.5	500	**16308.92**	**10690.48**	*park*	*park*	*park*	*park*	
6	slit	1	13.5	500	**3935.04**	**14983.28**	*park*	*park*	*park*	*park*	
7	width	1	6	800	**9718.39**	**4399.96**	*park*	*park*	*park*	*park*	
8	width	1	6	950	**9643.39**	**4474.96**	*park*	*park*	*park*	*park*	
9	width	1	6	1100	**9568.39**	**4549.96**	*park*	*park*	*park*	*park*	
10	width	1	6	1250	**9493.39**	**4624.96**	*park*	*park*	*park*	*park*	
11	width	1	6	1400	**9418.39**	**4699.96**	*park*	*park*	*park*	*park*	
12	slit	2	3	500	*park*	*park*	**8092.36**	**230.38**	*park*	*park*	
13	slit	2	4.5	500	*park*	*park*	**11037.14**	**3175.16**	*park*	*park*	

Step	Scan	Pair	λ (Å)	Width (μs)	**1a** Skf5	**1b** Skf1	**2a** Skf6	**2b** Skf2	**3a** Skf3	**3b** Skf4	Run no.
14	slit	2	6	500	*park*	*park*	**13981.19**	**6119.21**	*park*	*park*	
15	slit	2	8	500	*park*	*park*	**1239.05**	**10043.73**	*park*	*park*	
16	slit	2	10.5	500	*park*	*park*	**6143.81**	**14948.49**	*park*	*park*	
17	slit	2	13.5	500	*park*	*park*	**12029.10**	**4167.12**	*park*	*park*	
18	width	2	6	800	*park*	*park*	**13831.19**	**6269.21**	*park*	*park*	
19	width	2	6	950	*park*	*park*	**13756.19**	**6344.21**	*park*	*park*	
20	width	2	6	1100	*park*	*park*	**13681.19**	**6419.21**	*park*	*park*	
21	width	2	6	1250	*park*	*park*	**13606.19**	**6494.21**	*park*	*park*	
22	width	2	6	1400	*park*	*park*	**13531.19**	**6569.21**	*park*	*park*	
23	slit	3	3	500	*park*	*park*	*park*	*park*	**10459.75**	**150.78**	
24	slit	3	4.5	500	*park*	*park*	*park*	*park*	**14063.40**	**3754.43**	
25	slit	3	6	500	*park*	*park*	*park*	*park*	**999.65**	**7357.35**	
26	slit	3	8	500	*park*	*park*	*park*	*park*	**5802.67**	**12160.37**	
27	slit	3	10.5	500	*park*	*park*	*park*	*park*	**11805.55**	**1496.58**	
28	slit	3	13.5	500	*park*	*park*	*park*	*park*	**2341.92**	**8699.62**	
29	width	3	6	800	*park*	*park*	*park*	*park*	**849.65**	**7507.35**	
30	width	3	6	950	*park*	*park*	*park*	*park*	**774.65**	**7582.35**	
31	width	3	6	1100	*park*	*park*	*park*	*park*	**699.65**	**7657.35**	
32	width	3	6	1250	*park*	*park*	*park*	*park*	**624.65**	**7732.35**	
33	width	3	6	1400	*park*	*park*	*park*	*park*	**549.65**	**7807.35**	

B Derivations

B.1 Bias of the single-edge method and its effect on the band

Write $v = h/(m_n\lambda)$. The neutrons that define the edge pass the disc at x_{true} and reach the monitor at $T = \bar{t}_0 + (x_{\text{true}} - \bar{t}_0)\, L_{\text{mon}}/L_i$ (Eq. 6 with $d_0 = 0$). The method computes $x_{\text{assumed}} = L_i T/L_{\text{mon}} = x_{\text{true}} - \bar{t}_0(1 - L_i/L_{\text{mon}})$. This is Eq. 7. The offset is then too large by the same amount: $\hat{\delta}_i = \delta_i + \bar{t}_0(1 - L_i/L_{\text{mon}})$.

`calc6Phases` commands the edge for a target λ_* at $L_i\lambda_*/v$, assuming departure at T0. With $\hat{\delta}_i$ in place of δ_i, the edge actually crosses the beam at $L_i\lambda_* m_n/h + \bar{t}_0(1 - L_i/L_{\text{mon}})$. The neutrons it meets left at about $\bar{t}_0$, so their wavelength is

$$\lambda_{\text{edge}} = \frac{h}{m_n}\,\frac{L_i\lambda_* m_n/h + \bar{t}_0(1 - L_i/L_{\text{mon}}) - \bar{t}_0}{L_i} = \lambda_* - \frac{h}{m_n}\frac{\bar{t}_0}{L_{\text{mon}}},$$

the same for every disc. This is Eq. 8. The error-function bias adds a disc-dependent part on top.

B.2 The area theorem

Let disc a's transmission rise from 0 to 1 between $o - \tau/2$ and $o + \tau/2$ (any monotonic shape $r(t)$ with $r(o) = \frac{1}{2}$ and $r(o-u) + r(o+u) = 1$, e.g. linear). Let disc b's fall in the mirror way around c. If $c - o \geq \tau$ the two ramps do not overlap. The product of the gates is r_a on the rising part, 1 in between and r_b on the falling part. Its integral is

$$\int g_a g_b\, dt = \int_{o-\tau/2}^{o+\tau/2} r_a\, dt + (c - o - \tau) + \int_{c-\tau/2}^{c+\tau/2} r_b\, dt = \tfrac{\tau}{2} + (c - o - \tau) + \tfrac{\tau}{2} = c - o.$$

A spread of departure times convolves this with a normalized distribution, which leaves the integral unchanged. The number of neutrons through the slit is therefore $(c - o)$ times the flux per microsecond. The only correction is the slow variation of that flux across the slit, which the model handles.

C Software and reproduction

All scripts are in `Topic_2026_choppers/` (Research ORNL repository); run them with the Anaconda Python.

- `chopper_model.py`: geometry, departure-time model, gates, a port of `hexaSub.c`.
- `calibration_sim.py`: the simulated instrument, both methods, the width scan. `python calibration_sim.py --trials 3` (baseline); `--k-true 1 --tag sens_k1`; `--tau-true 250 --tag sens_tau250`.
- `report_figures.py`: Figs. 1–5, 6, 7, 13, 14.
- `report_tables.py`: the result tables, read from the CSV files the simulation writes.
- The tool (Section 10) is its own git repository, `slitcal` (`slitcal.py`, `slitcal_gui.py`, `README.md`); `analysis/slitcal_test/` holds the synthetic archive, its truth and the tool's output.
- Source copies and hashes: `source/` and `SOURCE.md`.

D Notation

λ	neutron wavelength (Å)
h/m_n	3956.0346 m Å/s
L_i, L_{mon}, L_{det}	distance from the moderator to disc i, monitor (10.006 m), detector (18.12 m)
P_i	commanded phase (`TotalDelay`) of disc i
δ_i	phase offset of disc i (Eq. 1)
t_0, $\bar{t}_0$	departure time from the moderator, and its mean
d_0	DAS time zero
τ_b	time for a disc edge to sweep across the beam (367 μs)
w	slit width: time between disc a opening and disc b closing